# CHAPTER 67 Liver

The liver is the largest of the abdominal viscera, occupying a substantial portion of the upper abdominal cavity. It occupies most of the right hypochondrium and epigastrium, and frequently extends into the left hypochondrium as far as the left anterior axillary line (Fig. 67.1). As the body grows from infancy to adulthood, the liver rapidly increases in size. This period of growth reaches a plateau around 18 years and is followed by a gradual decrease in liver weight from middle age. The ratio of liver to body weight decreases with growth from infancy to adulthood. The liver weight is 4–5% of body weight in infancy and decreases to approximately 2% in adulthood. The size of the liver also varies according to sex, being smaller in females, and body size, enlarging with fat deposition. It has an overall wedge shape, which is, in part, determined by the form of the upper abdominal cavity into which it grows. The narrow end of the wedge lies towards the left hypochondrium, and the anterior edge points anteriorly and inferiorly. The superior and right lateral aspects are shaped by the anterolateral abdominal and chest wall, as well as the diaphragm. The inferior aspect is shaped by the adjacent viscera. The capsule is no longer thought to play an important part in maintaining the shape of the liver; it is notable that it allows expansion when the liver hypertrophies in response to disease, surgical resection or contralateral embolization of the portal vein or hepatic artery.

Throughout life, the liver is reddish brown in colour, although this can vary, depending on the fat content. Obesity is the most common cause of excess fat in the liver (steatosis); the liver assumes a more yellowish tinge as its fat content increases and gains a bluish tinge with venous obstruction. The texture of the organ is usually soft to firm, although it depends partly on the volume of blood it contains and on its fat and fibrous tissue content.

The liver performs a wide range of metabolic activities required for homeostasis, nutrition and immune defence. For example, it is important in the removal and breakdown of toxic, or potentially toxic, materials from the blood; the regulation of blood glucose and lipids; the storage of certain vitamins, iron and other micronutrients; the synthesis of proteins and clotting factors; the metabolism of amino acids; and bile production. It is involved in a plethora of other biochemical reactions. Since the majority of these processes are exothermic, a substantial part of the thermal energy production of the body, especially at rest, is provided by the liver. The liver is populated by phagocytic macrophages, components of the mononuclear phagocyte system capable of removing particulates from the blood stream. It is an important site of haemopoiesis in the fetus.

An account of the more common eponyms relating to the anatomy and surgery of the liver is provided in Stringer (2009).

## SURFACES OF THE LIVER

The liver is usually described as having superior, anterior, right, posterior and inferior surfaces, and has a distinct inferior border (Figs 67.2–67.3). However, the superior, anterior and right surfaces are continuous and no definable borders separate them. It is more appropriate to group them as the diaphragmatic surface, which is mostly separated from the inferior, or visceral, surface by a narrow inferior border. At the infrasternal angle, the inferior border is adjacent to the anterior abdominal wall and accessible to examination by percussion, but not usually palpable, except on deep inspiration. In the midline, the inferior border of the liver is near the transpyloric plane. In women and children, the border often projects a little below the right costal margin.

**Superior surface** The superior surface is the largest and lies immediately below the diaphragm, separated from it by peritoneum, except for a small triangular area where the two layers of the falciform ligament diverge. The majority of the superior surface lies beneath the right dome, but there is a shallow cardiac impression centrally that corresponds to the position of the heart above the central tendon of the diaphragm. The left side of the superior surface lies beneath part of the left dome of the diaphragm.

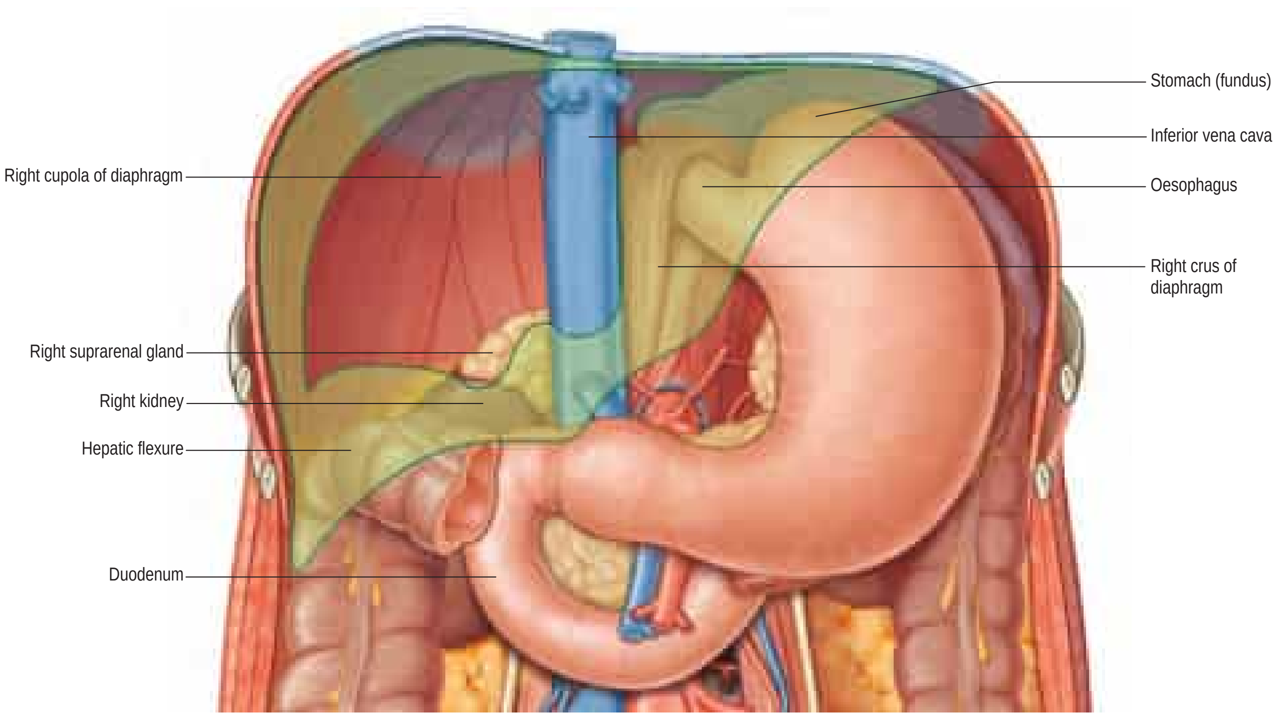


**Fig. 67.1** The 'bed' of the liver. The outline of the liver is shaded green. The central bare area is unshaded.

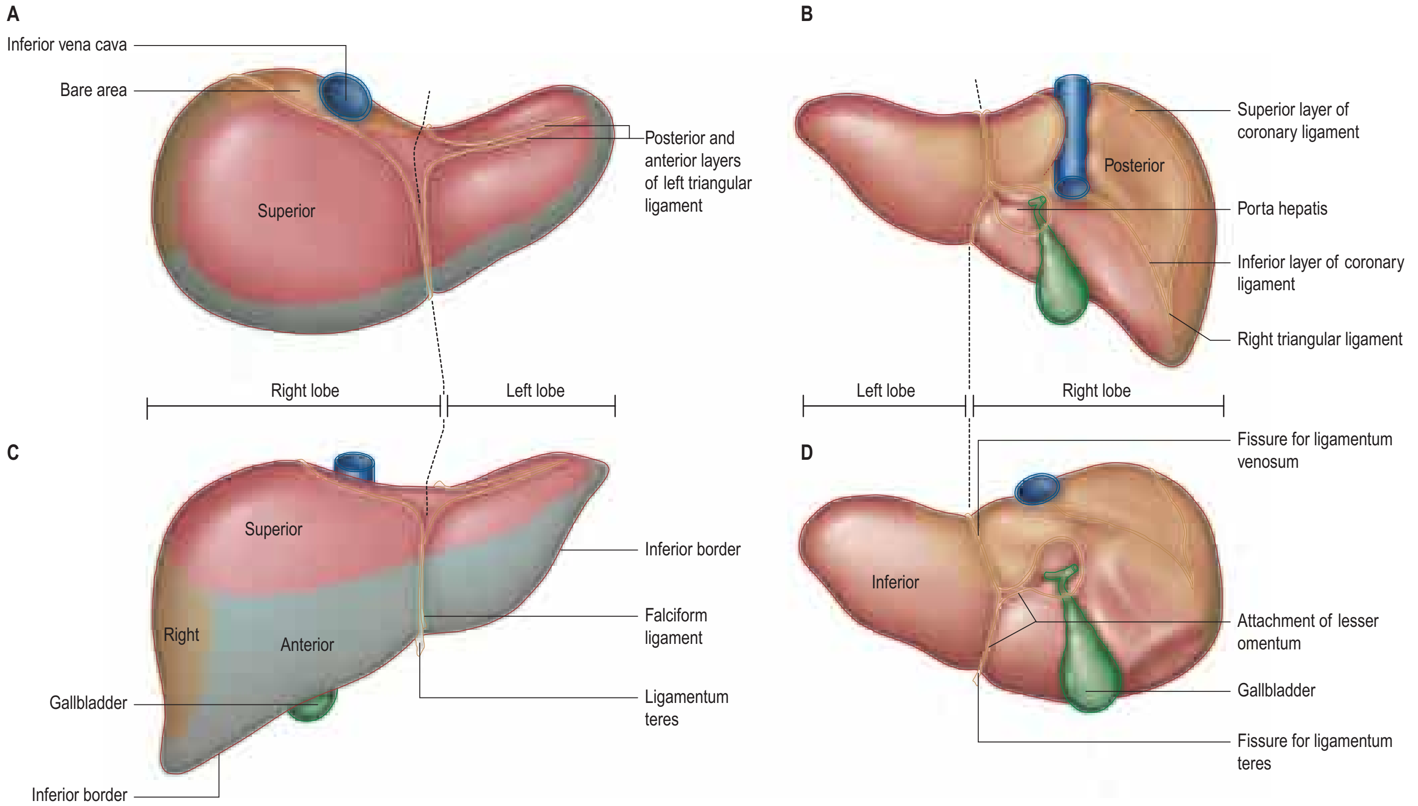


**Fig. 67.2** The surfaces and external features of the liver. **A**, Superior view. **B**, Posterior view. **C**, Anterior view. **D**, Inferior view.

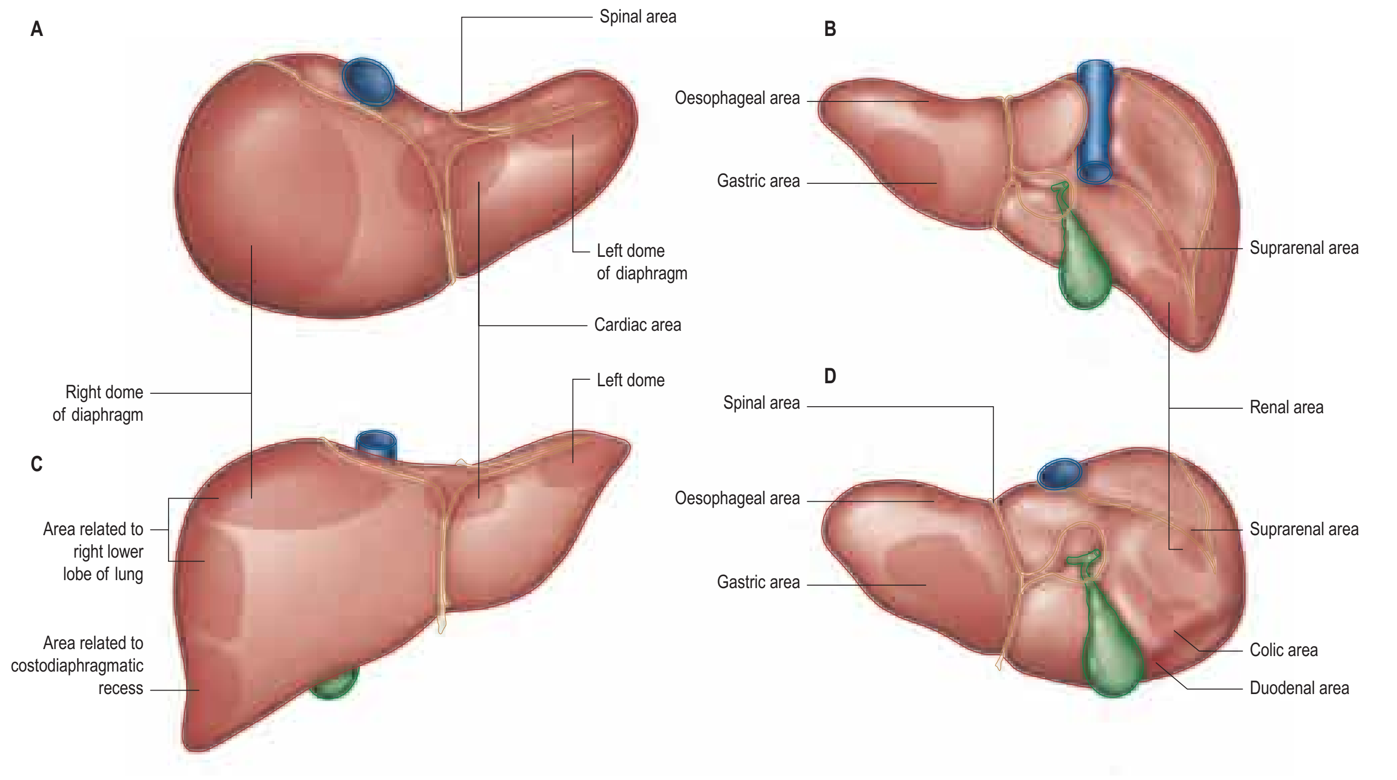


**Fig. 67.3** Relations of the liver. **A**, Superior view. **B**, Posterior view. **C**, Anterior view. **D**, Inferior view.

**Anterior surface** The anterior surface is approximately triangular and convex, and is covered by peritoneum, except at the attachment of the falciform ligament. Much of it is in contact with the anterior attachment of the diaphragm. On the right, the diaphragm separates it from the pleura and sixth to tenth ribs and cartilages, and on the left, from the seventh and eighth costal cartilages.

**Right surface** The right surface is covered by peritoneum and lies adjacent to the right dome of the diaphragm, which separates it from the right lung and pleura and the seventh to eleventh ribs. The right lung and basal pleura lie above and lateral to its upper third, between the diaphragm and the seventh and eighth ribs. The diaphragm, the costodiaphragmatic recess lined by pleura, and the ninth and tenth ribs lie lateral to the middle and lower thirds of the right surface. Lateral to the lower third, the diaphragm and thoracic wall are in direct contact. Rarely, the hepatic flexure and proximal transverse colon lie on a long mesentery over the right, anterior and superior surfaces of the liver; this may lead to misdiagnosis of pneumoperitoneum on a radiograph or very rarely cause symptoms (Chilaiditi's syndrome) (see p. 1144).

**Posterior surface** The posterior surface is convex, wide on the right but narrow on the left. A median concavity corresponds to the forward convexity of the vertebral column close to the attachment of the ligamentum venosum. Much of the posterior surface is attached to the diaphragm by loose connective tissue, forming the triangular 'bare area'. The inferior vena cava lies in a groove or tunnel in the medial end of the 'bare area'. To the left of the caval groove, the posterior surface of the liver is formed by the caudate lobe, and covered by a layer of peritoneum continuous with that of the inferior layer of the coronary ligament and the layers of the lesser omentum. The caudate lobe is related to the diaphragmatic crura and the right inferior phrenic artery above the aortic hiatus, and separated by these structures from the descending thoracic aorta.

The fissure for the ligamentum venosum separates the caudate lobe from the left lobe. The fissure cuts deeply in front of the caudate lobe and contains the two layers of the lesser omentum. The posterior surface of the left lobe bears a shallow impression near the upper end of the fissure for the ligamentum venosum that is caused by the abdominal oesophagus. The posterior surface of the left lobe to the left of this impression is related to the fundus of the stomach. Together, these posterior relations make up what is sometimes referred to as the 'bed' of the liver (see Fig. 67.1).

**Inferior surface** The inferior surface is bounded by the inferior edge of the liver. It blends with the posterior surface in the region of the origin of the lesser omentum, the porta hepatis and the inferior layer of the coronary ligament, and is marked near the midline by a sharp fissure that contains the ligamentum teres (the obliterated fetal left umbilical vein). The gallbladder usually lies in a shallow fossa but this is variable; it may have a short mesentery or be completely intrahepatic and lie within a cleft in the liver parenchyma. The quadrate lobe lies between the fissure for the ligamentum teres and the gallbladder.

The inferior surface of the left lobe is related inferiorly to the fundus of the stomach and the upper lesser omentum. The quadrate lobe lies adjacent to the pylorus, the first part of the duodenum and the lower part of the lesser omentum. Occasionally, the transverse colon lies between the duodenum and the quadrate lobe. To the right of the gallbladder, the inferior surface is related to the hepatic flexure of the colon, the right suprarenal gland and right kidney, and the first part of the duodenum (see Fig. 67.1).

## SUPPORTS OF THE LIVER

The liver is stabilized and maintained in its position in the right upper quadrant of the abdomen by both static and dynamic factors. A three-tier classification of supporting structures has been proposed: the suspensory attachments at the posterior abdominal wall to the inferior vena cava, hepatic veins, coronary and triangular ligaments (primary factors); the support provided by the right kidney, right colic flexure and duodenopancreatic complex (secondary factors); and the attachment to the anterior abdominal wall and diaphragm by the falciform ligament (tertiary factors) (Flament et al 1982).

The surgical implications of these different factors are important for understanding the pathophysiology of blunt liver trauma and when considering the stability of transplanted liver grafts. The inferior vena cava and the hepatic veins, especially the right hepatic vein, appear to be the most important anatomical structures that support the bulk of the liver. Other factors that influence the position of the liver within the abdominal cavity include positive intra-abdominal pressure and the movement of the diaphragm during respiration.

## GROSS ANATOMICAL LOBES

Historically, the liver has been divided on the basis of its external appearance into right, left, caudate and quadrate lobes, which are, in part, defined by peritoneal ligamentous attachments. Additional liver lobes have been reported but are rare.

**Right lobe** The right lobe is the largest in volume and contributes to all surfaces of the liver. It is divided from the left lobe by the falciform ligament anteriorly and superiorly and the ligamentum venosum and fissure for the ligamentum teres inferiorly. On the inferior surface, to the right of the grooves formed by the ligamentum teres and ligamentum venosum, there are two prominences separated by the porta hepatis; the caudate lobe lies posterior, and the quadrate lobe lies anterior. The gallbladder lies in a shallow fossa to the right of the quadrate lobe.

**Left lobe** The left lobe is the smaller of the two main lobes, although it is nearly as large as the right lobe in young children. It lies to the left of the falciform ligament with no subdivisions. It is substantially thinner than the right lobe, having a thin apex that points into the left upper quadrant.

**Quadrate lobe** The quadrate lobe is visible as a prominence on the inferior surface of the liver, to the right of the groove formed by the ligamentum teres (and thus is incorrectly said to arise from the right lobe, although it is functionally related to the left hemi-liver). It lies anterior to the porta hepatis and is bounded by the gallbladder fossa to the right, a short portion of the inferior border anteriorly, the fissure for the ligamentum teres to the left, and the porta hepatis posteriorly. Like the caudate lobe, its morphology varies between individuals (Joshi et al 2009).

**Caudate lobe** The caudate lobe is visible as a prominence on the inferior and posterior surfaces to the right of the groove formed by the ligamentum venosum; it lies posterior to the porta hepatis. To its right is the groove for the inferior vena cava. Above, it continues into the superior surface on the right of the upper end of the fissure for the ligamentum venosum. In gross anatomical descriptions, this lobe is said to arise from the right lobe but it is functionally separate.

## FUNCTIONAL ANATOMICAL DIVISIONS

Current understanding of the functional anatomy of the liver is based on Couinaud's division of the liver into eight (subsequently nine, then later revised back to eight) functional segments, based on the distribution of portal venous branches in the parenchyma (Couinaud 1957). Further understanding of the intrahepatic biliary anatomy, especially of the right ductal system, was enhanced by contributions from Hjortsjö (1951) and Healey and Schroy (1953), who used the biliary system as the main guide for division of the liver (**Fig. 67.4**).

The liver is divided into four portal sectors by the four main branches of the portal vein. These are right lateral, right medial, left medial and left lateral (sometimes, the term posterior is used in place of lateral, and anterior in place of medial). The three main hepatic veins lie between these sectors as intersectoral veins. These intersectoral planes are also called portal fissures (or scissures). Each sector is subdivided into segments (usually two), based on their supply by tertiary divisions of the vascular biliary (Glissonian) sheaths.

### Fissures of the liver

Knowledge of the fissures of the liver is essential for understanding liver surgery. Three major fissures (main, left and right portal fissures), not visible on the surface, run through the liver parenchyma and contain the three main hepatic veins. Three minor fissures (umbilical, venous and fissure of Gans) are visible as physical clefts of the liver surface. The fissure of Gans is also known as Rouvière's sulcus or the incisura hepatis dextra. Accessory fissures are rare.

**Main portal fissure** The main fissure, sometimes called Cantlie's line, extends from the midpoint of the gallbladder fossa at the inferior margin of the liver back to the midpoint of the inferior vena cava, and contains the middle hepatic vein. It separates the liver into right and left hemi-livers. Segments V and VIII lie immediately to the right, and segment IV immediately to the left, of the fissure.

**Left portal fissure** The left portal fissure divides the left hemi-liver into medial (anterior) and lateral (posterior) sectors. It extends from the midpoint of the inferior edge of the liver between the falciform ligament and the left triangular ligament to the point that marks the confluence of the left and middle hepatic veins. It contains the left hepatic vein and separates the left medial (anterior) and left lateral (posterior) sectors; segment III lies immediately anteriorly and segment II posteriorly.

**Right portal fissure** The right portal fissure divides the right hemi-liver into lateral (posterior) and medial (anterior) sectors. The plane of the right fissure is the most variable of the portal fissures and runs approximately diagonally through the anatomical right lobe from the lateral end of the inferior border to the termination of the right hepatic vein. The fissure divides the right anterior sector to its left (segments V and VIII) from the right posterior sector to its right (segments VI and VII), and contains the right hepatic vein. The right fissure traverses the thickest portion of liver parenchyma that is commonly transected during liver resection.

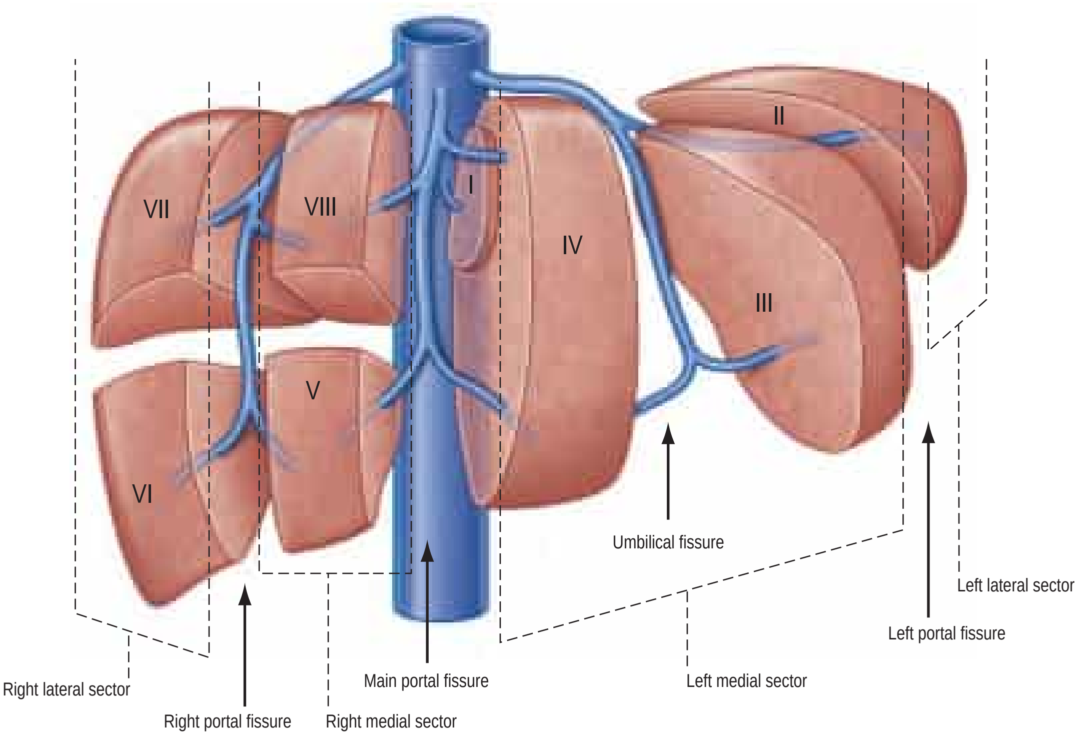


**Fig. 67.4** The fissures and sectors of the liver. (Right lateral = right posterior; right medial = right anterior.)

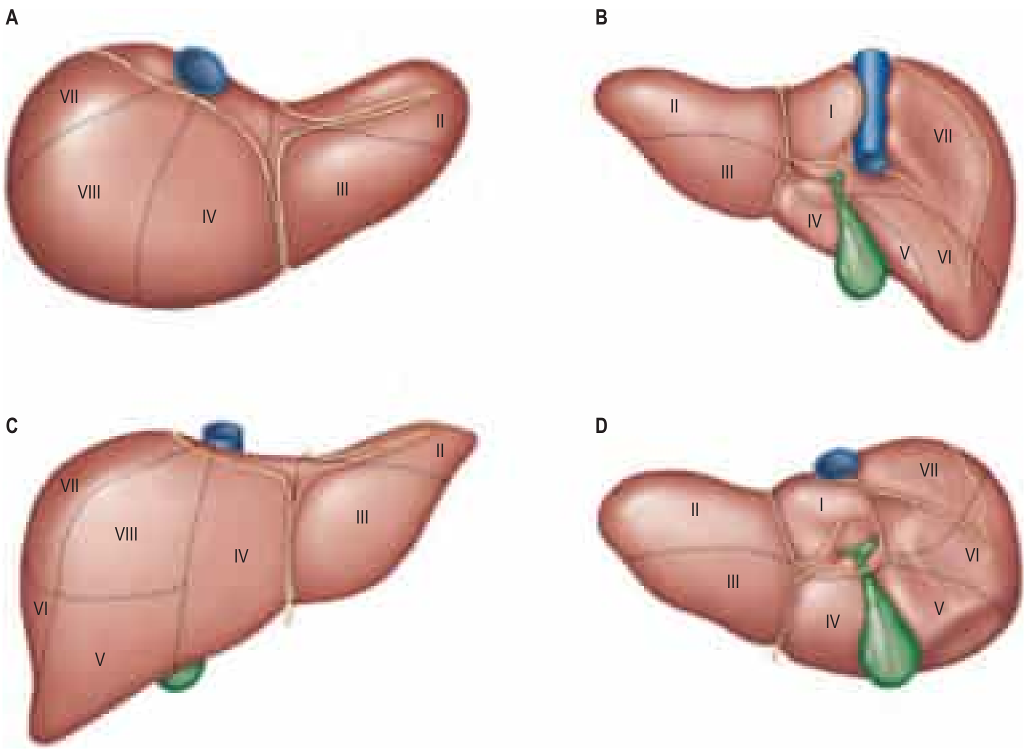


**Fig. 67.5** Segments of the liver (after Couinaud). **A**, Superior view. **B**, Posterior view. **C**, Anterior view. **D**, Inferior view. The segments are usually referred to by number (name): I (caudate); II (left lateral superior); III (left medial inferior); IV (left medial superior) (sometimes subdivided into superior and inferior parts); V (right medial inferior); VI (right lateral inferior); VII (right lateral superior); VIII (right medial superior).

**Umbilical fissure** The umbilical fissure (or fissure for the ligamentum teres) separates segment III from segment IV within the left medial sector and contains a major branch of the left hepatic vein (the umbilical fissure vein). It is marked anteriorly by the attachment of the falciform ligament and inferiorly by the ligamentum teres, where it may be covered by a bridge of liver tissue extending between segments III and IV. This liver bridge is usually avascular and can be divided safely with diathermy during surgery. The umbilical fissure also contains the umbilical portion of the left portal vein, segmental bile ducts converging to form the left hepatic duct, and the terminal branches of the left branch of the hepatic artery. The umbilical portion of the left portal vein offers direct surgical access to the left portal vein; this is important for mobilization of the left portal vein in surgery for hilar cholangiocarcinoma and in operations to restore intrahepatic portal blood flow after portal vein occlusion. A knowledge of the arrangement of the vascular and biliary structures within the umbilical fissure is also essential when splitting the liver for an adult and paediatric recipient and for live donor liver transplantation for a child recipient.

**Venous fissure** The venous fissure (or fissure for the ligamentum venosum) is in direct continuity with the umbilical fissure on the undersurface of the liver and contains the ligamentum venosum (the obliterated ductus venosus). It lies between the caudate lobe and segment II.

**Fissure of Gans** The fissure of Gans (or Rouvière's sulcus) lies on the undersurface of the right lobe of the liver behind the gallbladder fossa. It often marks the variable site of division of the portal pedicle to the right posterior sector.

## Sectors and segments of the liver

### Sectors

The sectors of the liver are made up of between one and three segments: right lateral sector = segments VI and VII; right medial sector = segments V and VIII; left medial sector = segments III and IV (and part of I); left lateral sector = segment II (Fig. 67.5). Segments are numbered

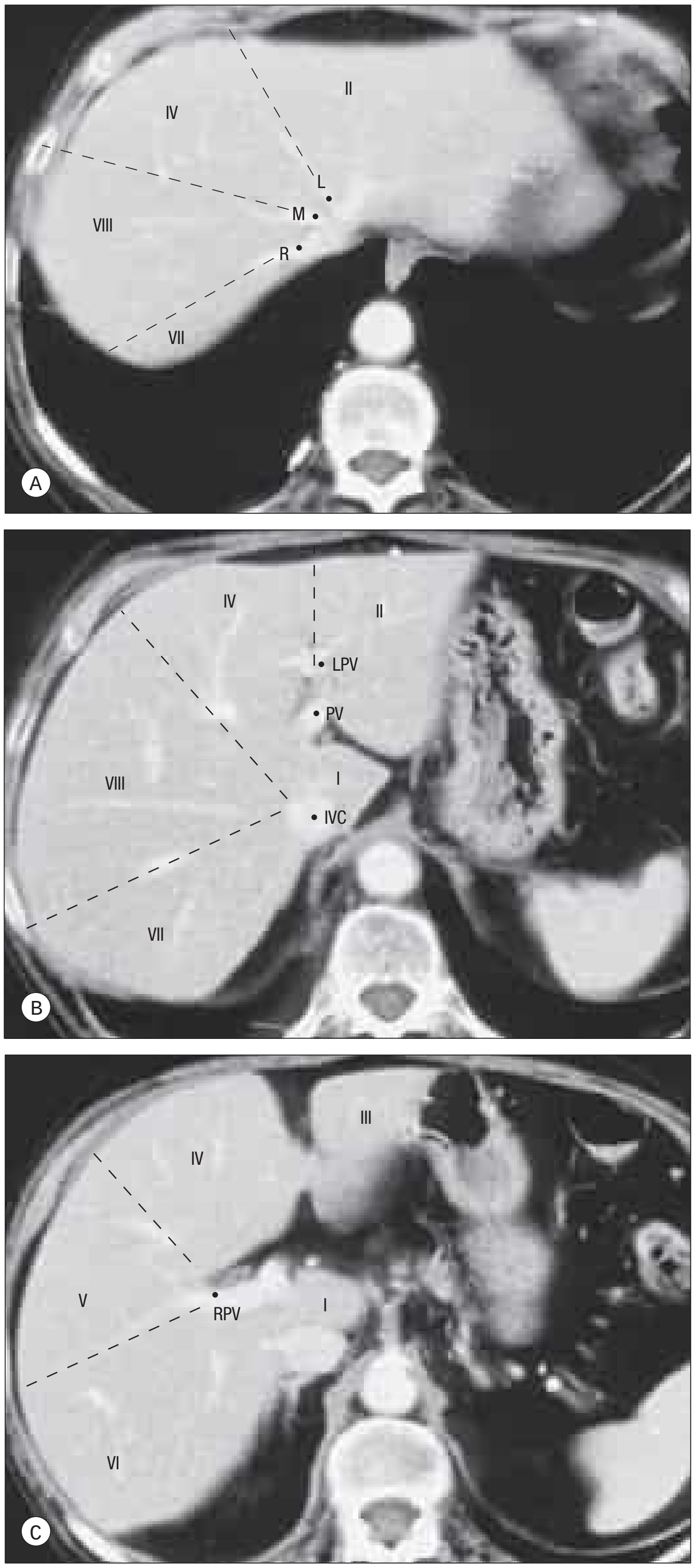


**Fig. 67.6** Segments of the liver seen on axial computed tomography (CT) scan. **A**, A contrast enhanced CT shows the left (L), middle (M) and right (R) hepatic veins at the superior aspect of the liver, marking the left, main and right portal fissures. **B**, Inferior to this, the caudate lobe (segment I) lies between the inferior vena cava (IVC) and the portal vein (PV). The left portal vein (LPV) separates segment II superiorly from segment III inferiorly. **C**, The right portal vein (RPV) divides segments V and VI inferiorly (**C**) from segments VII and VIII superiorly (**B**).

clockwise from below, starting with segment I and ending with segment VIII (see Figs 67.4–67.5; **Fig. 67.6**).

**Segment I** Segment I corresponds to the anatomical caudate lobe and lies posterior to segment IV, with its left half directly adjacent to segment II and its medial half wrapped around the retrohepatic segment of the inferior vena cava to a variable extent. In European literature, it is subdivided into three parts (caudate process, Spiegel lobe and paracaval portion), while in the Far East, right and left subdivisions are recognized. The Glissonian sheaths to segment I arise from both right and left main sheaths; the segment therefore receives vessels from both the left and right branches of the portal vein and hepatic arteries. The venous drainage of the caudate lobe is directly into the inferior vena cava by multiple small tributaries that nearly always arise from the lower, and occasionally from the middle, third of the segment, but rarely from the upper third. The bile ducts draining the segment are closely related to the confluence of the right and left hepatic ducts (see Chapter 68), such that excision of bile duct tumours affecting the hilum of the liver usually requires removal of segment I.

**Segment II** Segment II lies posterolateral to the left portal fissure and is the only segment in the left lateral sector of the liver. It often has only one Glissonian sheath and drains into the left hepatic vein. Rarely, a separate vein drains directly into the inferior vena cava.

**Segment III** Segment III lies between the umbilical fissure and the left portal fissure, and is supplied by one to three Glissonian sheaths; it drains into the left hepatic vein. The vein of the falciform ligament can provide an alternative drainage route for segment III. Very rarely, the venous drainage is to the middle hepatic vein and reconstruction is required following a right hepatectomy involving resection of the middle hepatic vein and in split liver transplantation (Dar et al 2008).

**Segment IV** Segment IV lies between the umbilical fissure and the main portal fissure, immediately anterior to segment I. Segment IV is supplied by three to five Glissonian sheaths, most of which arise in the umbilical fissure; their origins are often close to those that supply segments II and III. Occasionally, segment IV is supplied by branches from the main left pedicle. The main venous drainage of segment IV is into the middle hepatic vein but the segment can also drain into the left hepatic vein through the vein of the falciform ligament. Segment IV has been divided into IVa superiorly and IVb inferiorly; this is relevant to transverse hepatectomy, in which segments IVb, V and VI are removed along a transverse portal plane (Sugarbaker 1990).

**Segment V** Segment V is the inferior segment of the right medial sector and lies between the middle and the right hepatic veins. Its size is variable, as are the number of Glissonian sheaths that supply it from the right anterior sheath. Venous drainage is usually into the right and middle hepatic veins, but may be direct into the inferior vena cava via an inferior right hepatic vein.

**Segment VI** Segment VI forms the inferior part of the right lateral sector posterior to the right portal fissure. It is often supplied by two to three branches from the right posterior Glissonian sheath, and occasionally the Glissonian sheath to segment VI can arise directly from the right pedicle. Venous drainage is normally into the right hepatic vein but, like in segment V, may be via an inferior right hepatic vein directly into the inferior vena cava.

**Segment VII** Segment VII forms the superior part of the right lateral sector and lies behind the right hepatic vein. The sheath to segment VII is often single. The venous drainage is into the right hepatic vein; occasionally, the segment can drain directly into the inferior vena cava.

**Segment VIII** Segment VIII is the superior part of the right medial sector. The right medial (anterior) sectoral sheath ends in segment VIII and supplies it, after giving off branches to segment V. Venous drainage is to the right and middle hepatic veins.

## Peritoneal attachments and ligaments of the liver

The liver is attached to the anterior abdominal wall, diaphragm and other viscera by several peritoneal ligaments.

### *Falciform ligament and ligamentum teres*

The liver is attached to the anterior abdominal wall by the falciform ligament, derived from the ventral mesogastrium in the embryo. The two layers of this ligament pass posteriorly and slightly to the right from the posterior surface of the anterior abdominal wall and the undersurface of the diaphragm in the midline, and attach to the anterior and superior surfaces of the liver. On the dome of the superior surface, the right leaf runs laterally and is continuous with the upper layer of the coronary ligament. The left layer of the falciform ligament turns medially and is continuous with the anterior layer of the left triangular ligament. The ligamentum teres is the obliterated remnant of the left umbilical vein of the fetus; it runs in the lower free border of the falciform ligament and continues into a fissure on the inferior surface of the liver. In fetal life, the left umbilical vein opens into the left portal vein. It is supposed to be obliterated in adult life but frequently remains partially patent; in reality, its lumen is usually closed rather than obliterated and it can be dissected at the umbilicus, dilated and used to access the left portal vein in more than half of individuals. The ligamentum teres may reopen in conditions such as portal hypertension, when it forms a sizeable collateral channel.

The ligamentum teres is important in abdominal surgery for several reasons. It is often divided in upper abdominal surgery to optimize access to the upper abdominal viscera or as the first step in mobilization of the liver. It is vascularized by numerous small arterial branches (mainly from the artery to segment IV) and para-umbilical veins, and these anastomose with branches of the superior epigastric artery; it is therefore important to ligate or coagulate the ligament during its division. The ligamentum teres is a guide to the segment III hepatic duct in hepaticojejunostomy formation, and to the left portal vein lying in the umbilical fissure during mesenterico-portal bypass (Vellar et al 1998, di Francesco et al 2014).

### Coronary ligament

The coronary ligament is formed by the reflection of the peritoneum from the diaphragm on to the superior and posterior surfaces of the right lobe of the liver. A large triangular area of liver devoid of peritoneal covering, the so-called 'bare area' of the liver, lies between the two layers of the coronary ligament. Here, the liver is attached to the diaphragm by areolar tissue, which is in continuity inferiorly with the anterior pararenal space. On the right, the two layers of the coronary ligament converge laterally to form the right triangular ligament. On the left, the two layers become closely applied, and form the left triangular ligament. The upper layer of the coronary ligament is reflected superiorly on to the inferior surface of the diaphragm and inferiorly on to the right and superior surfaces of the liver. The lower layer of the coronary ligament is reflected inferiorly over the right suprarenal gland and right kidney, and superiorly on to the inferior surface of the liver. Surgical division of the right triangular and coronary ligaments allows the right lobe of the liver to be brought forwards, and exposes the lateral aspect of the inferior vena cava behind the liver.

### Triangular ligaments

The left triangular ligament is a double layer of peritoneum that extends for a variable length over the superior border of the left lobe of the liver. Medially, the anterior leaf is continuous with the left layer of the falciform ligament, and the posterior layer is continuous with the left layer of the lesser omentum. The left triangular ligament lies in front of the abdominal part of the oesophagus and part of the fundus of the stomach. Division of the left triangular ligament allows the left lobe of the liver to be mobilized for exposure of the abdominal oesophagus and crura of the diaphragm. The left triangular ligament is an important stabilizing factor for the left lobe after excision of the right lobe of the liver. Its division will result in the left lobe being unstable, to the extent that it can rotate and displace into the space created under the right hemidiaphragm, which, in turn, can compromise the venous outflow of the left lobe with consequent liver dysfunction. If it is divided, the left lobe should be stabilized by reattaching the falciform ligament to the anterior abdominal wall.

The right triangular ligament is a short structure that lies at the right lateral limit of the 'bare area' of the liver and represents the convergence laterally of the two layers of the coronary ligament.

### Lesser omentum

The lesser omentum is a double layer of peritoneum that extends from the lesser curvature of the stomach and proximal duodenum to the inferior surface of the liver. Its free margin contains the portal triad (Ch. 63). Its attachment to the inferior surface of the liver is L-shaped. The vertical component follows the line of the fissure for the ligamentum venosum, the fibrous remnant of the ductus venosus. More inferiorly, the attachment runs horizontally to complete the L in the porta hepatis. At its upper end, the left layer of the lesser omentum is continuous with the posterior layer of the left triangular ligament, and the right layer is continuous with the coronary ligament as it encloses the inferior vena cava. At its lower end, the two layers of the lesser omentum diverge to surround the structures of the porta hepatis. A thin, fibrous condensation of fascia usually runs from the medial end of the porta hepatis into the fissure for the ligamentum teres. This fascia is continuous with the lower border of the falciform ligament where the ligamentum teres re-emerges at the inferior border of the liver. Care should be taken when dividing the lesser omentum because an aberrant or accessory left hepatic artery may run within it; when present, it invariably extends to the liver at the base of the umbilical fissure, and may be identified by a pulsation in the lesser omentum close to the umbilical fissure.

### Ligamentum venosum

The ligamentum venosum represents the obliterated venous connection that existed in the fetus between the left branch of the portal vein and the termination of the left hepatic vein (the ductus venosus). It is used as a guide to gain control of the left hepatic vein outside the liver during surgery. By dividing the ligament close to its insertion into the left hepatic vein and retracting it laterally, the angle between the left and the middle hepatic veins may be accessed.

This can be helpful in left liver resection with preservation of the caudate lobe, but if the caudate lobe is to be removed, the left hepatic vein (or, more usually, the confluence of the left and middle hepatic veins with the inferior vena cava) can be approached more posteriorly without division of the ligamentum venosum.

## Porta hepatis, hepatoduodenal ligament and hilar plate

The porta hepatis is a deep transverse fissure on the inferior surface of the liver. It is situated between the quadrate lobe anteriorly and the caudate process posteriorly, and contains the portal vein, hepatic artery and hepatic nervous plexuses as they ascend into the parenchyma of the liver, and the right and left hepatic ducts and some lymph vessels that emerge from the liver. The hepatic ducts usually lie anterior to the portal vein and its branches, and the hepatic artery with its branches lies between the two (Figs 67.7–67.8). However, the right hepatic artery sometimes lies anterior to the common hepatic duct; this variation is important during bile duct reconstruction by hepaticojejunostomy

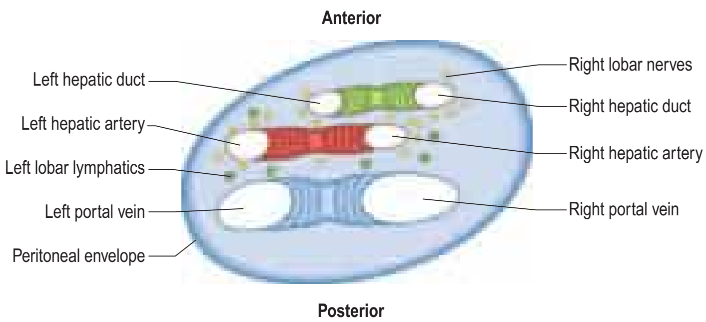


**Fig. 67.7** A cross-section of the structures at the porta hepatis viewed from above.

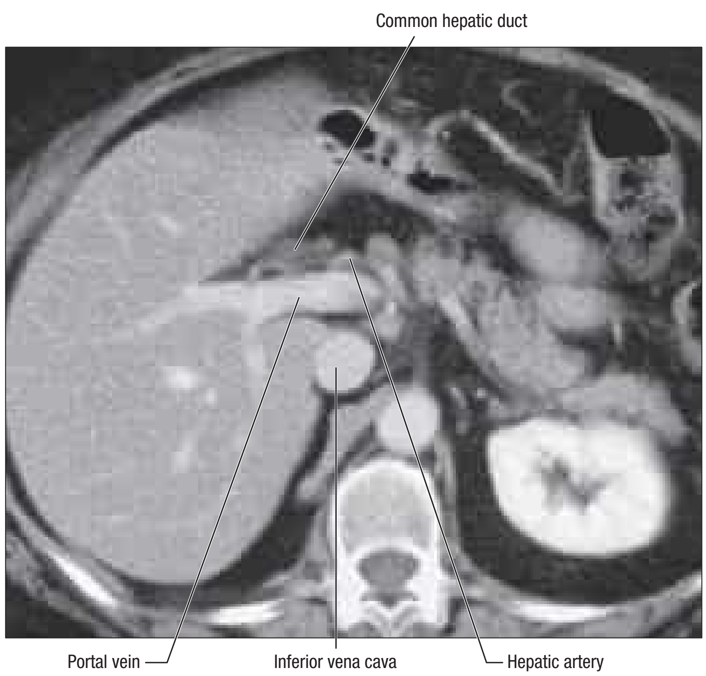


**Fig. 67.8** An axial CT of the porta hepatis. The hepatic ducts lie anteriorly, the portal vein posteriorly, and hepatic artery between the two.

(Michels 1966). All these structures are enveloped within a perivascular sheath of loose connective tissue that surrounds the vessels and bile ducts as they course through the liver parenchyma, and is continuous with the fibrous hepatic capsule (of Glisson). The dense aggregation of vessels, supporting connective tissue, and liver parenchyma just above the porta hepatis is often referred to as the 'hilar plate' of the liver. Understanding the concept of the hilar plate is important in surgical approaches to the hilar structures; division or lowering of the hilar plate is essential for optimum surgical access to the left hepatic duct.

The hepatic artery, bile duct and portal vein extend between the porta hepatis and the upper border of the duodenum in the free edge of the hepatoduodenal ligament, which forms the anterior boundary of the epiploic foramen. Rapid control of the vessels entering the porta hepatis (the hepatic pedicle) can be obtained by compressing them in the free edge of the lesser omentum (a 'Pringle' manœuvre); this is conveniently done by dividing the lesser omentum immediately to the left of these structures and passing a tape around them through the epiploic foramen.

The left hepatic duct is extrahepatic as it descends obliquely along the undersurface of segment IV (the quadrate lobe) to the confluence of the hepatic ducts. Access to this extrahepatic segment of the left hepatic duct is particularly useful when performing a biliary-enteric bypass procedure to treat a stricture of the hepatic duct confluence. The right hepatic duct is more intrahepatic and extension of a hepatico-jejunal anastomosis on to the right hepatic duct necessitates incision of the liver parenchyma in the gallbladder fossa.

### Glissonian sheaths

Glisson's capsule of the liver becomes condensed as Glissonian sheaths around the branches of the portal triad structures as they enter the liver parenchyma and subdivide into segmental branches. Thus, each bile duct, hepatic artery and portal vein is surrounded by a single fibrous sheath, which Couinaud called the 'Valoean sheath' (after Valoeus, an anatomist from the Middle Ages who first described the liver capsule). Within each sheath, the portal vein is surrounded by loose areolar connective tissue, making dissection of the portal vein relatively easy. The fibrous tissue around the bile ducts and hepatic artery is tougher, and dissection of these structures is more difficult. This sheath arrangement facilitates surgical control of the right and left vasculobiliary pedicles of the liver, as well as sectoral and segmental divisions in complex liver resections (Yamamoto et al 2012).

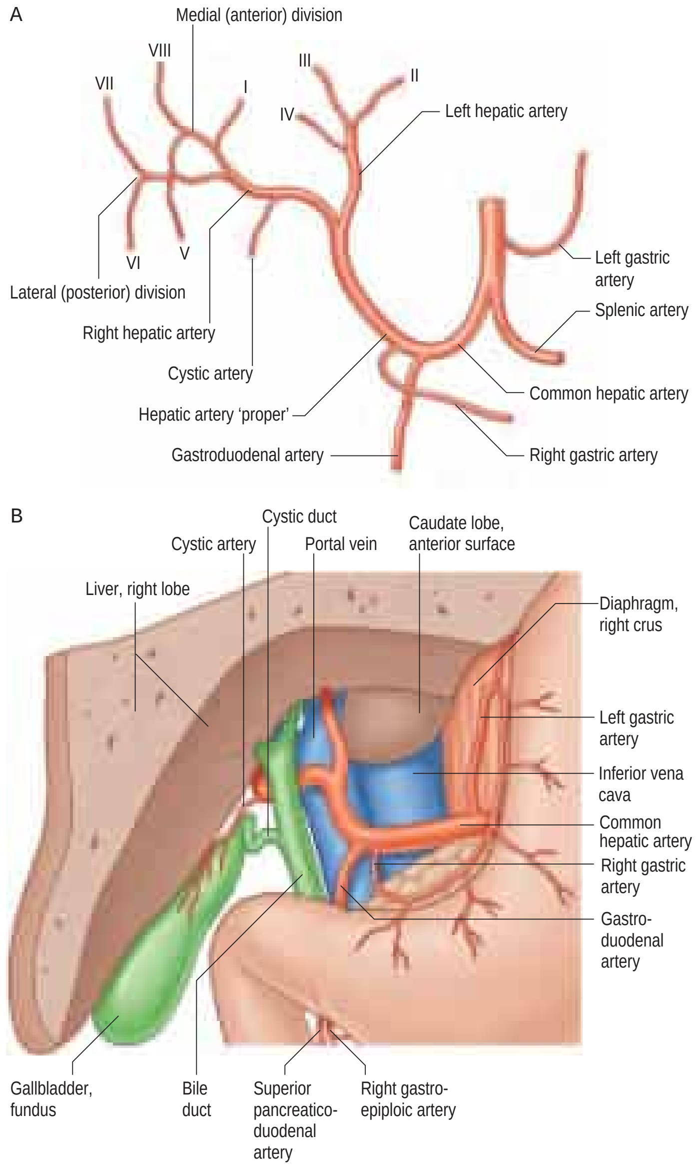


**Fig. 67.9** The hepatic artery. **A**, Branches. **B**, Usual relations of the hepatic artery, bile duct and portal vein to each other in the lesser omentum: anterior aspect, portion of the liver removed.

## VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

The blood vessels connected with the liver are the portal vein, hepatic artery and hepatic veins. The portal vein and hepatic artery ascend in the lesser omentum to the porta hepatis, where each usually bifurcates. The common hepatic duct and lymphatic vessels descend from the porta hepatis alongside the portal vein and hepatic artery (see Fig. 67.8). The hepatic veins leave the liver via its posterior surface and run directly into the inferior vena cava.

### Hepatic artery

In adults, the common hepatic artery is intermediate in size between the left gastric and splenic arteries. In fetal and early postnatal life, it is the largest branch of the coeliac trunk. The hepatic artery gives off the right gastric and gastroduodenal arteries, as well as branches to the bile duct and gallbladder from its right hepatic branch (**Fig. 67.9A**). After originating from the coeliac trunk, the hepatic artery passes anteriorly and laterally above the upper border of the pancreas to the upper aspect of the first part of the duodenum. It is subdivided into the common hepatic artery, from the coeliac trunk to the origin of the gastroduodenal artery, and the 'hepatic artery proper', from that point to its bifurcation. It ascends anterior to the portal vein and medial to the bile duct within the free margin of the lesser omentum in the anterior wall of the epiploic foramen. It divides into right and left branches at a variable level below the porta hepatis. The right branch of the hepatic artery usually crosses posterior (occasionally anterior) to the common hepatic duct (**Fig. 67.9B**). This close proximity often means that the right hepatic artery is involved in bile duct cancer earlier than the left hepatic artery. Occasionally, the right hepatic artery crosses anterior to the common hepatic duct and is more vulnerable to injury in biliary surgery. It almost always divides into an anterior branch supplying segments V and VIII, and a posterior branch supplying segments VI and VII. The anterior division also often supplies a branch to segment I and the gallbladder. The hepatic artery proper sometimes divides at a low level close to its origin and, occasionally, it divides at a higher level, well to the left of the porta hepatis. The main significance of an early division is that the right branch may pass behind the portal vein (Lanouis and Jamieson 1993). The segmental arteries of the liver are macroscopically end arteries, although some collateral circulation occurs between segments via fine terminal branches.

Anatomical variants of the normal arrangement of the hepatic artery are found in about one-third of individuals and are important to recognize because they are relevant to surgical and interventional radiological procedures (Covey et al 2002, Michels 1966, López-Andújar et al 2007, Saba and Mallarini 2011). An artery that supplies part of the liver in addition to its normal artery is defined as an accessory artery. A replaced hepatic artery is an artery that does not originate from an orthodox position and provides the sole supply to that part of the liver.

The most common anatomical variants are a replaced or accessory left hepatic artery that arises from the left gastric artery, or a replaced or accessory right hepatic artery that arises from the superior mesenteric artery, both occurring in 10–20% of individuals.

Variations in the intrahepatic arteries are common and may be surgically important. For example, the segment IV artery most commonly arises from the left hepatic artery, but in up to 30% of cases, it arises from the right hepatic artery or the hepatic artery proper (Onishi et al 2000). The segment IV artery never arises to the right of the common hepatic duct; thus, if the right hepatic artery is divided to the right of the common hepatic duct, this arterial supply to segment IV is not endangered. Failure to recognize this variation may compromise the blood supply to segment IV following right hepatectomy, and is especially important during right lobe donation for live donor liver transplantation.

A replaced hepatic artery proper may arise from the superior mesenteric artery (**Fig. 67.10**) and can be suspected at surgery by a relatively superficial portal vein (reflecting the absence of a hepatic artery that would normally ascend in front of the vein). More commonly, a replaced or accessory right hepatic artery arises from the superior mesenteric artery (see Fig. 67.10; **Fig. 67.11**). In such cases, the variant artery runs in the lesser omentum behind the portal vein and bile duct, and can usually be identified at surgery by palpable pulsation behind the portal vein. An accessory right hepatic artery may be injured during resection of the pancreatic head because the artery lies in close proximity to the portal vein. Occasionally, a replaced or accessory left hepatic artery arises from the left gastric artery, entering the liver through the umbilical fissure; this artery provides a source of collateral supply in cases where the arteries at the porta hepatis are occluded, but it may also be injured during mobilization of the stomach, as it lies in the upper portion of the lesser omentum. Rarely, an accessory left or right hepatic artery may arise from the gastroduodenal artery or directly from the aorta. The presence of replaced arteries can be life-saving in patients with bile duct cancer; because they are further away from the bile duct they tend to be spared from infiltration by the cancer, making excision of the tumour feasible. Knowledge of these variations is also essential when performing whole and split liver transplantation.

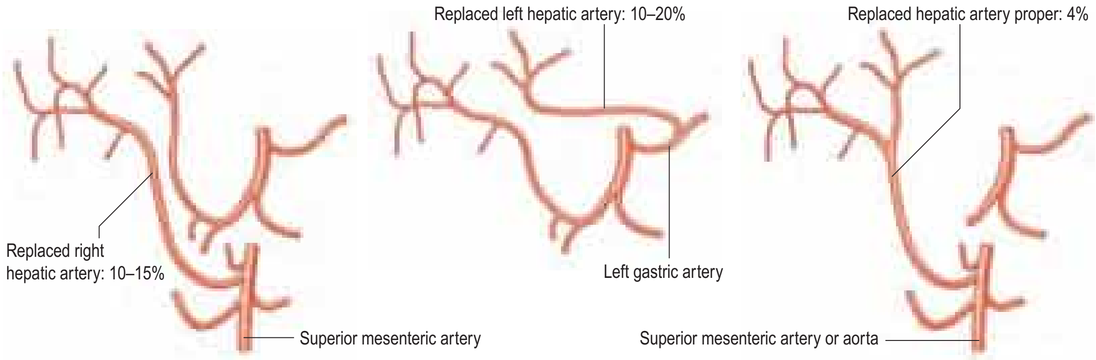


**Fig. 67.10** Common hepatic artery variants.

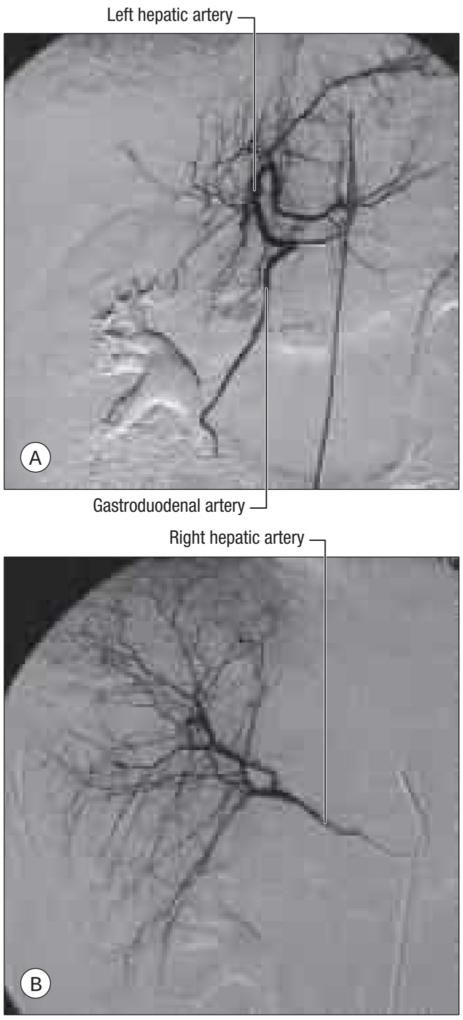


**Fig. 67.11** Hepatic arteriograms. **A**, A selective hepatic arteriogram shows normal left hepatic artery branches and small right hepatic artery branches. **B**, The right hepatic artery is arising from the origin of the superior mesenteric artery.

## Veins

The liver has two venous systems. The portal system conveys venous blood from the majority of the gastrointestinal tract and its associated organs to the liver (p. 1039). The hepatic venous system drains blood from the liver parenchyma into the inferior vena cava.

### *Portal vein*

The portal vein is formed behind the neck of the pancreas, usually from the convergence of the superior mesenteric and splenic veins (**Fig. 67.12**; see also Figs 70.8b, 59.8). Its origin lies in the transpyloric plane between the lower border of the body of the first lumbar vertebra and the upper border of the body of the second lumbar vertebra (Mirjalili et al 2012). The portal vein is approximately 8 cm long and ascends obliquely to the right behind the first part of the duodenum, the common bile duct and gastroduodenal artery, and anterior to the inferior vena cava. It enters the right border of the lesser omentum and ascends anterior to the epiploic foramen to reach the right end of the porta hepatis, where it divides into right and left main branches, which enter the liver. In the lesser omentum, the portal vein lies posterior to both the bile duct and the hepatic artery. It is surrounded by the hepatic nerve plexus and accompanied by numerous lymphatics and some lymph nodes. The portal vein contains smooth muscle in its wall and, in experimental animals at least, has well-developed spontaneous contractions with frequencies between 0.01 and 1 Hz (Burt 2003). It is typically valveless.


**Fig. 67.12** A coronal CT of the portal vein and superior mesenteric vein.

The main extrahepatic tributaries of the portal vein are the left gastric (coronary) vein, which ends in the left margin of the portal vein, and the posterior superior pancreaticoduodenal vein near the head of the pancreas. The portal vein divides into right and left branches at the hilum of the liver (**Fig. 67.13**). The left portal vein has a longer extrahepatic course (4–5 cm) than the right portal vein, tends to lie more horizontal, and is often smaller in calibre. It has a horizontal portion that runs along the inferior surface of segment IV and invariably gives branches to segment I and sometimes to segment IV. The left branch of the portal vein continues within the liver, giving off a segment II branch laterally before taking a more anterior and vertical course in the umbilical fissure. Here, it gives off branches to segments III and IV, and receives the obliterated left umbilical vein (ligamentum teres). The majority of the portal venous supply to segment IV comes from the left portal vein, and only occasionally from the right branch of the portal vein or its branches to segment V or VIII. The right branch of the portal vein is only 2–3 cm in length and usually divides into a right medial (anterior) sectoral division supplying segments V and VIII, and a right lateral (posterior) sectoral division supplying segments VI and VII. The medial division may give a branch to segment I.

Portal vein variations usually involve the right branch (Covey et al 2004). If the latter is absent, which occurs in about 10–15% of livers, the portal vein usually trifurcates into left portal, right medial and right lateral sectoral veins. This has implications for split liver and live donor liver transplantation, where its presence might be considered as a relative contraindication. The right lateral sectoral portal vein may arise from the portal vein, or the right medial sectoral portal vein may originate from the left portal vein, a variant that it is important to remember during left-sided liver resection. Rarely, the portal bifurcation is absent, in which case the portal vein enters the liver, giving off the right sectoral branches, and then turns left to supply the left lobe, which presents an added complexity in major liver surgery (Chaib 2009).

#### Porto-systemic shunts

Increased pressure within the portal venous system may result in dilation of portal venous tributaries and reversed flow at sites of porto-systemic anastomoses. Common sites of porto-systemic shunts are listed in **Table 67.1**.

### *Hepatic veins*

The liver drains by three major hepatic veins into the suprahepatic part of the inferior vena cava and via numerous minor hepatic veins that drain into the retrohepatic inferior vena cava. The adult retrohepatic inferior vena cava is 6–7 cm long and surrounded, to a variable extent, by segment I (Camargo et al 1996). The three major veins are located between the four sectors of the liver (see Fig. 67.6A; **Figs 67.14–67.15**). Thus, the right hepatic vein lies between the right medial and lateral sectors, the middle hepatic vein lies between the right and left hemilivers, and the left hepatic vein lies between the left medial and lateral sectors. During hepatic resection, the surgeon should transect the liver parenchyma slightly to the left or right of the particular fissure that is being opened to avoid the main trunk of a hepatic vein.

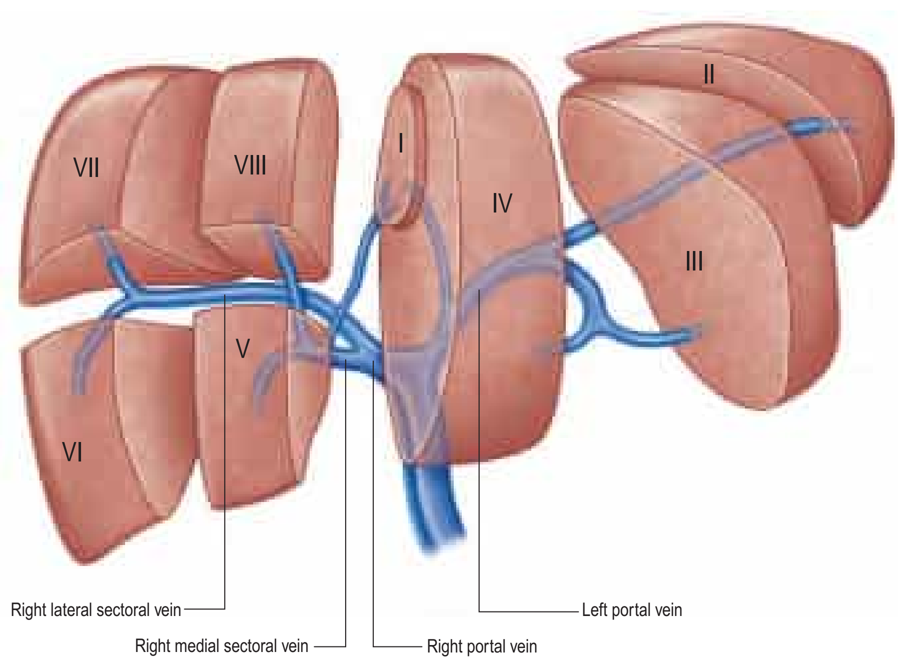


**Fig. 67.13** The main portal vein and its intrahepatic branches. (Right lateral = right posterior; right medial = right anterior.)

CHAPTER 67

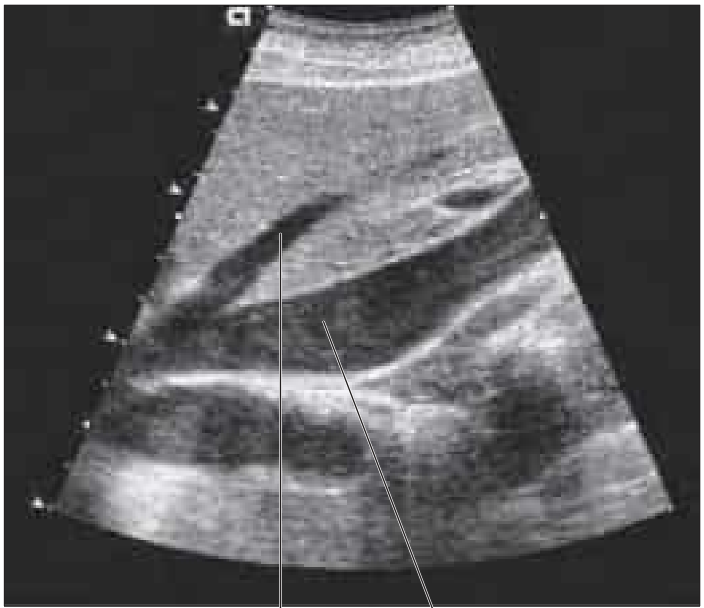


**Fig. 67.14** A sagittal ultrasound of the middle hepatic vein. The middle hepatic vein is seen draining into the inferior vena cava.

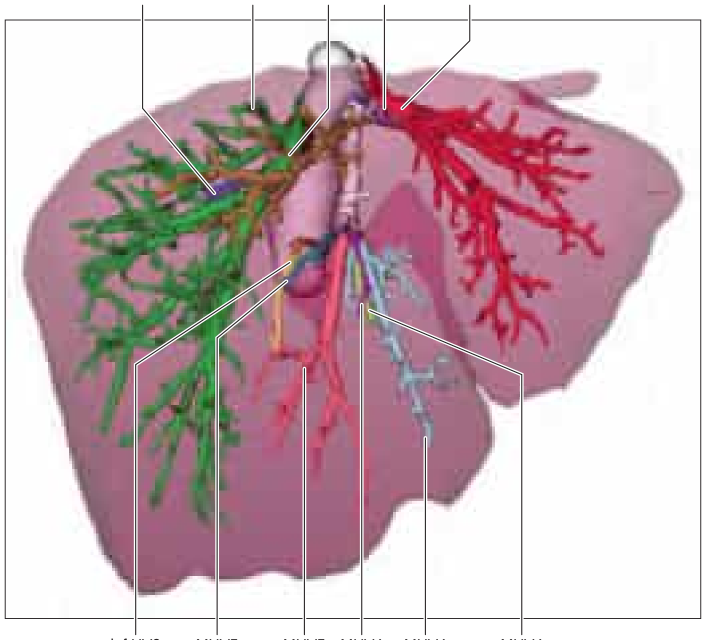


**Fig. 67.15** Modern CT analysis techniques are helpful in demonstrating variations in hepatic venous anatomy and this is very useful for planning live donor surgery for liver transplantation. Abbreviations: inf. HV1, inf. HV2, right inferior hepatic veins; LHV, left hepatic vein; LHV4, left hepatic vein branch to segment 4; MHV4, middle hepatic vein branches to segment 4; MHV5, middle hepatic vein branches to segment 5; MHV8, middle hepatic vein branches to segment 8; RHV, right hepatic vein. (Courtesy of MeVis Medical Solutions AG, Bremen, Germany.)

**Table 67.1 Common sites of porto-systemic anastomoses in portal hypertension and associated clinical implications**

| Portal vein tributaries | Systemic veins | Clinical presentations |
|---|---|---|
| Left gastric vein | Distal oesophageal veins draining into azygos and hemiazygos veins | Oesophageal and gastric varices |
| Superior rectal veins | Middle and inferior rectal veins draining into internal iliac and pudendal veins | Rectal varices |
| Persistent tributaries of left branch of portal vein in ligamentum teres | Periumbilical branches of epigastric and intercostal veins | 'Caput medusae' |
| Tributaries of right branch of portal vein overlying 'bare area' of liver | Retroperitoneal veins draining into azygos, hemiazygos, lumbar, intercostal and phrenic veins | Dilated retroperitoneal veins at risk during surgery or interventional procedures |
| Omental and colonic veins near hepatic and splenic flexures | Retroperitoneal veins near hepatic and splenic flexures | May be problematic during surgery |

## Right hepatic vein

This is the longest and largest hepatic vein. It is usually single, but occasionally remains as two trunks until it terminates by draining into the inferior vena cava. The right hepatic vein runs in the right portal fissure between the right medial and lateral sectors. It drains the whole of segments VI and VII, and variable proportions of segments V and VIII, depending on the extent to which these segments drain into the middle hepatic vein. The right hepatic vein is formed anteriorly near the inferior border of the liver and lies in a coronal plane through most of its course. It drains into the inferior vena cava near the upper border of the caudate lobe. Of the three major hepatic veins, the right hepatic vein is most variable in its size, not only due to the variable contribution of the middle hepatic vein to the drainage of segments V and VIII but also due to the existence of an accessory right inferior (30%) and/or middle hepatic vein (10%) (Fang et al 2012). The hepatic venous anatomy is particularly important in right lobe living donation and can be assessed preoperatively by computed tomography (CT) or magnetic resonance imaging (MRI) and/or by intraoperative ultrasound.

## Middle hepatic vein

The middle hepatic vein lies in the main portal fissure between the right and left hemi-livers. It usually joins the left hepatic vein and terminates in the inferior vena cava as a short common trunk (about 5 mm long); it ends as a single trunk in the inferior vena cava in fewer than 10% of individuals. The middle hepatic vein drains the central part of the liver and receives constant tributaries from segments IV, V and VIII. The vein from segment IV lies in a sagittal plane and enters the middle hepatic vein on its left side. The vein from segment VIII runs transversely into the right side of the middle hepatic vein and is sometimes large enough to be mistaken for the middle hepatic vein. Anteriorly, the middle hepatic vein drains some of segment V; the sizes of the tributaries draining segments V and VIII are variable. Intrahepatic venous anastomoses between the middle and right hepatic veins, particularly in segment VIII, have been reported in up to one third of adult livers (Hribernik and Trotovšek 2014).

## Left hepatic vein

The left hepatic vein lies between the left medial and left lateral sectors of the liver and drains segments II, III and, occasionally, IV. Small veins draining segment II and, occasionally, the superior part of segment IV may drain directly into the inferior vena cava. Usually, a major tributary of the left hepatic vein, the umbilical fissure vein, runs between segments III and IV and contributes to their drainage. Occasionally, the vein draining segment III ends separately in the confluence of the left and middle hepatic veins. These variations in venous drainage are of significance in split liver transplantation and live donor liver transplantation.

## Minor hepatic veins

Segment I veins drain directly into the inferior vena cava and vary in number from one to five. Since this segment has an independent venous drainage from the rest of the liver, in patients with Budd–Chiari syndrome, in which the major hepatic veins are blocked, segment I often continues to drain effectively and undergoes compensatory hypertrophy. There may be an accessory inferior or middle right hepatic vein, as well as several smaller 'retrohepatic' veins that drain the right lobe directly into the inferior vena cava. When present, they are of surgical importance, especially if greater than 5 mm in diameter; they drain segments V and VI independently of the three major hepatic veins and, therefore, a tumour involving the latter can be resected safely as long as venous drainage from the accessory veins is preserved. In live donor and split liver transplantation, larger accessory veins must be individually anastomosed to the recipient inferior vena cava to ensure adequate venous drainage.

## Transjugular intrahepatic porto-systemic shunt (TIPS) procedure for portal hypertension

Available with the *Gray's Anatomy* e-book

SECTION 8

In extreme cases of chronic portal hypertension, a large-calibre anastomosis between the portal and systemic circulations may be created within the liver parenchyma by inserting a stent between a large portal and hepatic vein within the liver. The stent is introduced through a catheter inserted into the internal jugular vein and guided into the liver under radiological control. This large shunt relieves the severe portal hypertension but tends to worsen incipient hepatic encephalopathy.

## Segmental anatomy of the liver in relation to hepatic resection

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## Lymphatic drainage

Lymph from the liver is rich in protein and is mostly a product of the hepatic sinusoids (Ohtani and Ohtani 2008). It passes, via deep and superficial pathways, to nodes above and below the diaphragm (Trutmann and Sasse 1994). Obstruction of hepatic venous drainage increases the flow of lymph in the thoracic duct.

### *Superficial hepatic lymphatics*

Superficial lymphatics run in subserosal areolar tissue over the surface of the liver and drain in four directions. Lymphatics from most of the posterior surface, including the caudate lobe, drain into nodes alongside the inferior vena cava; a few lymphatics from the posterior surface of the left lobe pass towards the oesophageal hiatus and nodes around the cardia. Lymphatics in the coronary and right triangular ligaments may pass directly to the thoracic duct. Lymphatics from most of the inferior, anterior and superior surfaces drain into hepatic nodes at the porta hepatis. A few lymphatics from the right superior surface accompany the inferior phrenic artery across the right diaphragmatic crus to drain into coeliac nodes.

### *Deep hepatic lymphatics*

Fine lymphatics within the portal triads and around interlobular veins merge to form larger vessels. Some ascend through the parenchyma to pass through the vena caval opening in the diaphragm and drain into inferior mediastinal nodes, but most drain to lymph nodes at the porta hepatis.

## INNERVATION

The liver has a dual innervation. The parenchyma is supplied by nerves arising from the hepatic plexus, which contains sympathetic and parasympathetic (vagal) fibres; they all enter the liver at the porta hepatis. The capsule is supplied by fine branches of the lower intercostal nerves, which also supply the parietal peritoneum, particularly around the 'bare area' and superior surface; distension or disruption of the liver capsule causes quite well-localized, sharp pain.

### *Hepatic plexus*

The hepatic plexus receives preganglionic parasympathetic fibres from the anterior and, to a lesser extent, the posterior vagus, and postganglionic sympathetic fibres via the coeliac and superior mesenteric plexuses. Nerve fibres accompany the branches of the portal triad into the liver, penetrating as far as the hepatocytes. Aminergic, cholinergic, peptidergic and nitrergic nerve fibres have been identified and appear to be involved in hepatic metabolism and the control of sinusoidal blood flow (McCuskey 2004); however, the transplanted, denervated liver indicates that this role is not essential. Multiple fine branches from the plexus supply the extrahepatic bile ducts and gallbladder; the vagal fibres are motor to the muscle of the bile ducts and gallbladder, and inhibitory to the sphincter of the bile duct. Branches from the plexus also run inferiorly with the right gastric artery to contribute to the supply of the pylorus; with the gastroduodenal artery and its branches to reach the pylorus, proximal duodenum and pancreas; and with the right gastroepiploic artery to provide a small contribution to the nerve supply of the stomach.

### *Referred pain*

Pain arising from the parenchyma of the liver is poorly localized. In common with other structures of foregut origin, pain is referred to the epigastrium. Stretch or irritation of the liver capsule by inflammation or neoplasia produces well-localized 'somatic' pain. Pathology involving the diaphragmatic surface of the liver may be referred via the phrenic nerve to the right shoulder region (C3,4,5 dermatomes).

## MICROSTRUCTURE

The liver is essentially an epithelial-mesenchymal outgrowth of the caudal part of the foregut, with which it remains connected via the biliary tree (see Ch. 60). Most of the surface of the liver is covered by a typical serosa, the visceral peritoneum. Beneath this, and enclosing the whole organ, is a thin (50–100 µm) capsule of connective tissue, from which extensions pass into the liver as septa and trabeculae. Branches of the hepatic artery and hepatic portal vein, together with bile ductules and ducts, run within these connective tissue trabeculae, which are termed portal tracts. The combination of the two types of vessel and a bile duct is termed a portal triad (Fig. 67.19); these

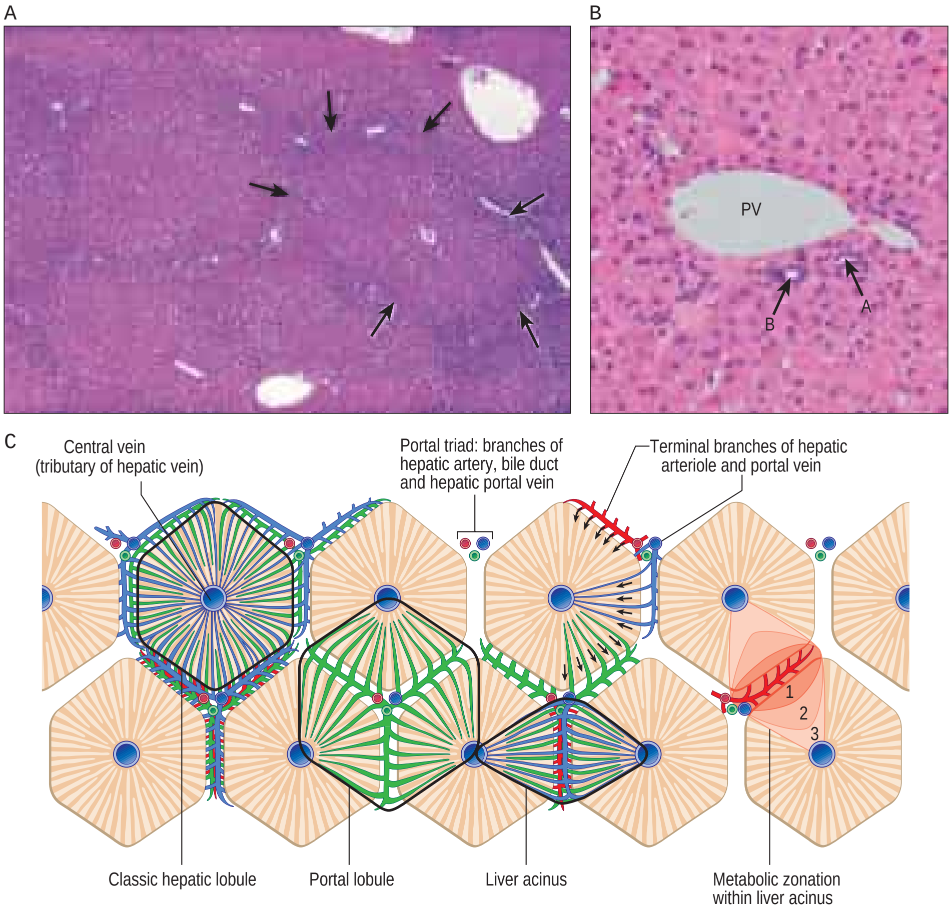



**Fig. 67.19** The structural organization of human liver tissue. **A**, Lobules bordered by delicate connective tissue septa (arrows), in which run branches of the hepatic portal vein, hepatic artery and bile duct, grouped as portal triads. A central vein drains each lobule. **B**, A portal triad containing branches of the hepatic portal vein (PV; generally the largest profile), the hepatic artery (A) and a bile duct (B), with typical round epithelial nuclei. Other small bile ductule and arteriolar branches are also visible. **C**, A functional view, in which the territories of the hepatic lobules are shown as regular hexagons (unlike their real appearance, which is highly variable). Functionally, the liver microarchitecture is better considered in terms of acini extending between adjacent central venules and divided into three zones as shown. (A, With permission from Dr JB Kerr, Monash University, from Kerr JB 1999 Atlas of Functional Histology. London: Mosby. B, Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

In 1888, Rex described the midline division of the liver, and in the 1950s Claude Couinaud's detailed description of the segmental anatomy of the liver formed the basis for subsequent advances in liver imaging and surgery. Goldsmith and Woodburne also published a detailed description of liver anatomy in 1957 but used a different terminology; confusion in hepatic nomenclature has persisted for several decades.

Couinaud divided the right and left hemi-livers into two *sectors* each, according to the secondary divisions of the portal vein. In the right hemi-liver, the anteromedial and posterolateral sectors were each subdivided into a superior and an inferior segment, producing a total of four segments. In the left hemi-liver, the medial sector was divided into two segments separated by the falciform ligament and ligamentum teres, while the lateral sector consisted of a single segment (II). The smallest segment, which Couinaud designated segment I, was located in the central and posterior part of the liver between the bifurcation of the portal vein and the inferior vena cava, and had independent hepatic vein and portal vein tributaries. Couinaud, therefore, described eight liver segments, numbered I to VIII: segments II, III and IV made up the left hemi-liver and segments V, VI, VII and VIII made up the right hemi-liver. In 1994, he added segment IX to describe the small region of the right hemi-liver lying adjacent to the right side of the inferior vena cava, but later recognized that this was better considered as part of segment I.

In 1982, Henri Bismuth used Couinaud's anatomical descriptions with some modifications as the basis for developing anatomical liver resections (Bismuth 2013). Couinaud's studies had been based on corrosion casts of injected cadaveric livers that had undergone slight postmortem flattening from resting on a firm surface. Bismuth renamed Couinaud's anteromedial and posterolateral sectors in the right hemi-liver as anterior and posterior, and also challenged Couinaud's subdivision of the left liver. Couinaud had described two sectors: a lateral sector consisting of a single segment, and a medial sector consisting of two segments separated by a branch of the left portal vein lying in continuity with the round ligament. Bismuth considered that this went against Couinaud's own description of individual segments containing a major branch of the portal vein, and suggested that the left medial sector also consisted of a single segment, i.e. segments III and IV were, in fact, 'half-segments' (Fig. 67.16). This scheme created a single sector containing two segments within the left hemi-liver. To avoid confusion, Bismuth chose to retain the segment numbers III and IV, albeit recognizing them as half-segments. The Bismuth classification recognized two hemi-livers, three sectors (right posterior, right anterior and left) and seven segments: segments 6 and 7 in the right posterior sector, segments 5 and 8 in the right anterior sector, half-segments 3 and 4 and segment 2 in the left sector, and segment 1 (numbering was changed from Roman to Arabic).

In 1986, Ken Takasaki described a different basis for subdividing the gross architecture of the liver, in which the portal vein had three branches (right, middle and left) and there were just two hepatic veins (right and middle, the left hepatic vein being a tributary of the middle hepatic vein; Fig. 67.17). Takasaki had divided the liver into three parts of almost equal volume, based on the three branches of the portal vein and the two hepatic veins. Apart from the terms used, Takasaki's description is similar to Bismuth's modification of Couinaud's anatomy in that the liver is considered to consist of three, and not four, territories. It fails to take into account the division of the left side of the liver by the falciform and round ligaments. The major advantage of Takasaki's description is the individualization of the middle part, facilitating the concept of central hepatectomy.

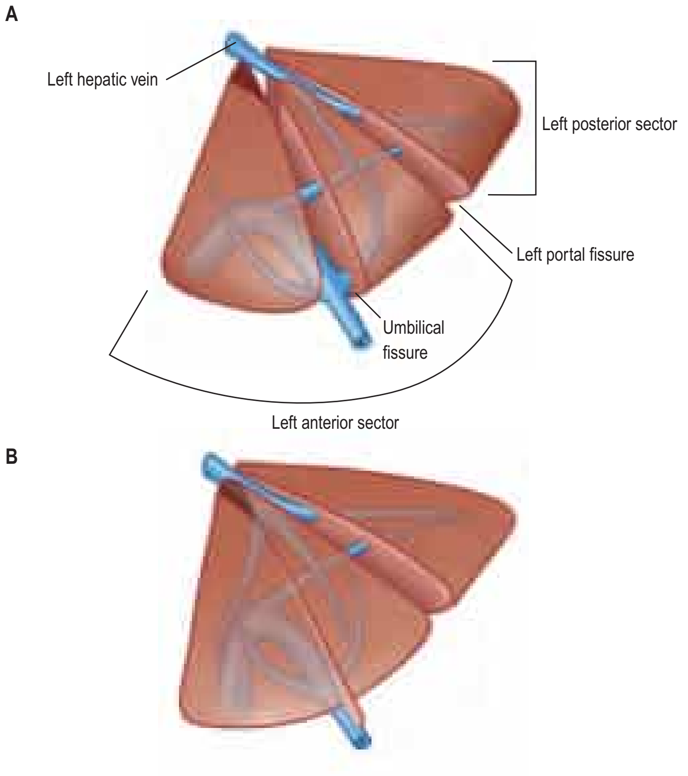



**Fig. 67.16** **A**, The umbilical fissure that divides the anterior sector of the left lobe into two segments is, in fact, an artificial scissure. **B**, When it is suppressed, the anterior sector appears as a single segment. (Redrawn with permission from Bismuth H. Surgical anatomy and anatomical surgery of the liver. World J Surg 1982;6:3–9, Springer.)

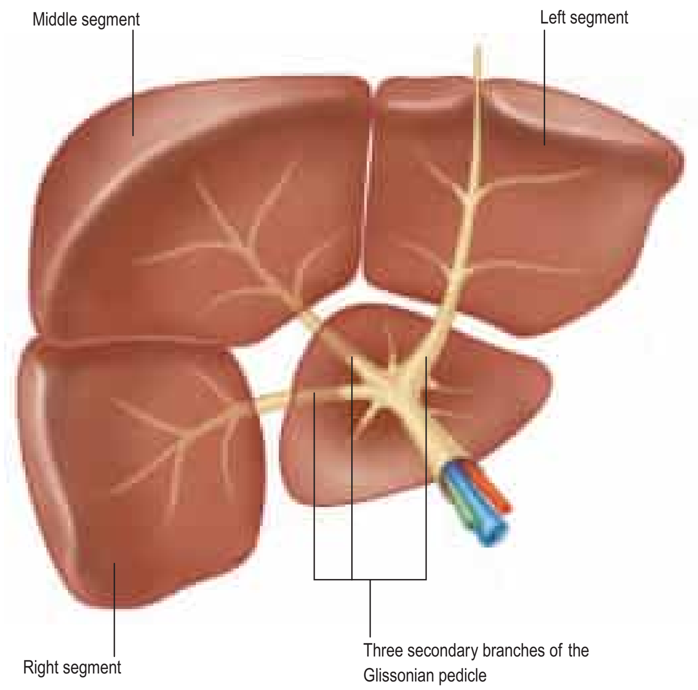



**Fig. 67.17** The liver is divided into three segments and a caudate area, according to the ramification of the Glissonian pedicles. (Redrawn based on Takasaki K, Koabayashi S, Tanaka S, et al 1986 Highly selected hepatic resection by Glissonian sheath-binding method. Dig Surg 3: 121.)

In 1998, the International Hepato-Pancreato-Biliary Association (IHPBA) developed an international committee on nomenclature of hepatic anatomy and resections, and this reported to the IHPBA General Assembly in Brisbane in 2000 (Terminology Committee of the IHPBA 2000). The chairman of the committee, Steven M. Strasberg, highlighted the confusion that arose from applying multiple terms to the same anatomical structure or operative procedure, and from using the same term to describe more than one structure or operation. For example, in 1997, there were 13 terms in use for the plane that defined the watershed between the right and left hemi-livers (called the midplane of the liver by the committee), and the terms 'lobe', 'lobectomy', 'segment' and 'segmentectomy' were being used to describe different parts of the liver and associated procedures, depending on whether the surgeon was European or North American. The anatomical basis for the Brisbane 2000 nomenclature was the usual branching pattern of the hepatic arteries and bile ducts within the liver. The ramification of the portal vein was said to follow an identical pattern in the right hemi-liver but to be different in the left hemi-liver. Three orders of branching resulted in successive division of the liver into two hemi-livers, four *sections* and eight segments. Watershed areas between territories subserved by branches of the hepatic artery and bile duct were demarcated by the midplane of the liver, a right and left intersectional plane, and several intersegmental planes. The terminology of liver resections was based directly on the anatomical terminology, and individual resections were termed hemihepatectomy (or hepatectomy), sectionectomy and segmentectomy; an extended resection of three sections was called a trisectionectomy (**Fig. 67.18**). There has been a clear trend towards adoption of this terminology since 2000 but it has not been universal, not least because the classification has some anatomical deficiencies. In particular, Couinaud's segments II and III became the left lateral section (which meant that removal became a left lateral sectionectomy) but this does not accurately represent the anatomical division. The introduction of the term 'section', as opposed to 'sector', was intended to introduce simpler terms for segments II and III (left lateral section) and segment IV (left medial section) that would mirror the arrangement in the right hemi-liver (right anterior and posterior sections).

Bismuth argued that classification of the lateral part of the left hemi-liver was particularly problematic. According to Couinaud, individual liver territories contain portal vein branches but are separated by the major hepatic veins. However, the lateral part of the left hemi-liver is demarcated by the falciform and round ligaments. Resection of this part of the liver remains one of the most commonly performed hepatectomies. In classical anatomy, the term lobe relates to part of an organ defined by an external fissure; removal of the part of the liver to the left of the falciform and round ligaments was originally called a left lobectomy. According to subsequent classifications, this resection has been termed a left lateral segmentectomy (Goldsmith and Woodburne 1957), bisegmentectomy (Couinaud 1957), resection of half a segment (Takasaki et al 1986), and left lateral sectionectomy (Terminology Committee of the IHPBA 2000). To avoid confusion, the term lobectomy is no longer used but Bismuth has argued for a return to classical anatomy and the use of the term left lobectomy. He has also argued that all hepatectomies should be defined by the specific segments that are resected. Thus, removal of a single segment should be described as 'segmentectomy n' where n is the specific number of the segment removed. Resection of two segments is a 'bisegmentectomy n+n1', and that of three segments a trisegmentectomy n+n1+n2 (Bismuth 2013).

Masatoshi Makuuchi, a member of the IHPBA Brisbane 2000 committee on nomenclature, has recently questioned the current status, pointing out that the differences in portal vein branching patterns in the left and right hemi-livers have been the major obstacle in consistently classifying liver territories to satisfy what is known about hepatic embryology, as well as descriptive and surgical anatomy. Makuuchi (2013) has commented that, in the embryo, the right and left hemi-livers develop synchronously and that the right and left umbilical veins initially join the right and left branches of the portal vein, respectively. At 4–5 weeks' gestation, the right umbilical vein is resorbed so that the volume of the left side of the liver increases compared to the right, but after occlusion of the left umbilical vein after birth, the left hemi-liver gradually shrinks and the right side enlarges. Makuuchi argues that the left lateral sector (segment II) and right lateral sector (segments VI and VII) were so named by Couinaud because the portal vein branches to both sectors are the first large branches arising from the left and right divisions of the portal vein. These two sectors also share the characteristic of having a major hepatic vein running on their medial border. This is contrary to the views of both Bismuth and Takasaki, who maintain that segment II should be amalgamated with segments III and IV into a single sector, and of the IHPBA Brisbane 2000 nomenclature committee, who considered segment IV to be a full sector. Makuuchi postulates that the left lateral sector shrinks after birth and, if segment IV is determined to be a sector, then the right medial sector of the liver should be demarcated by a plane that corresponds to the embryonic right umbilical vein. As the main portal pedicle should be located at the centre of the sector and not at its border, if the left and right hemi-livers are each divided into two sectors, it is more logical to regard segment II as a sector, as suggested by Couinaud, rather than segment IV.

Makuuchi also discussed the use of the term 'extended', as proposed by Bismuth and the IHPBA Brisbane 2000 nomenclature committee. He suggested that ambiguity could be eliminated by adopting the wording 'extended into segment X', where X is the specific segment number identified by Couinaud. Further revision of the nomenclature is likely in the future (Strasberg and Phillips 2013).

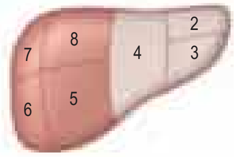
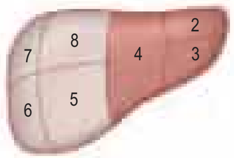
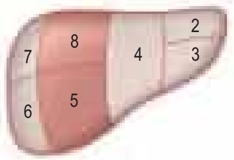
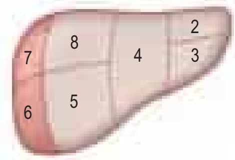
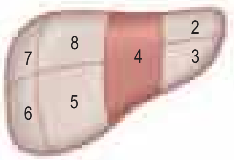
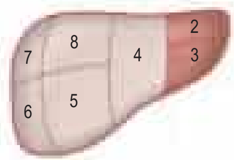
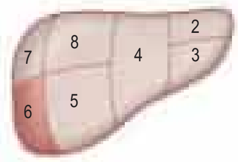
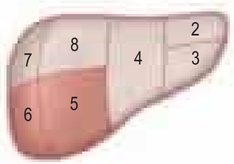
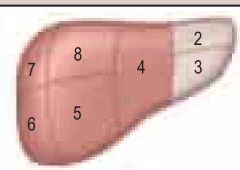
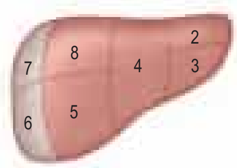
| Anatomical term | Couinaud segments | Term for surgical resection | Diagram (pertinent area is shaded) |
|---|---|---|---|
| Right liver or Hemi-liver | Sg 5-8 | Right hepatectomy or Right hemihepatectomy |  |
| Left liver or Hemi-liver | Sg 2-4 | Left hepatectomy or Left hemihepatectomy |  |
| Right anterior section | Sg 5,8 | Right anterior sectionectomy |  |
| Right posterior section | Sg 6,7 | Right posterior sectionectomy |  |
| Left medial section | Sg 4 | Left medial sectionectomy |  |
| Left lateral section | Sg 2,3 | Left lateral sectionectomy |  |
| Segments 1-8 | Any one of Sg 1-8 | Segmentectomy (e.g. segmentectomy 6) |  |
| Two contiguous segments | Any two of Sg 1-8 in continuity | Bisegmentectomy (e.g. bisegmentectomy 5,6) |  |
| **Extended resections** (Trisectionectomy) | | | |
| Sg 4-8 | | Right trisectionectomy (preferred term) or Extended right hepatectomy or Extended right hemihepatectomy |  |
| Sg 2,3,4,5,8 | | Left trisectionectomy (preferred term) or Extended left hepatectomy or Extended left hemihepatectomy |  |

**Fig. 67.18** Resectional terminology. (Redrawn with permission from Terminology Committee of the International Hepato-Pancreato-Biliary Association. The Brisbane 2000 terminology of liver anatomy and resections. HPB 2000;2:333–9.)

CHAPTER 67

structures are usually accompanied by one or more lymphatic vessels and nerves.

The liver parenchyma consists of a complex network of epithelial cells, supported by connective tissue, and perfused by a rich blood supply from the portal vein and hepatic artery. The epithelial cells, hepatocytes, carry out the major metabolic activities, but additional cell types possess storage, phagocytic and mechanically supportive functions. In the mature liver, hepatocytes are arranged mainly in plates (or cords when seen in two-dimensional sections) that are usually only one cell thick and are separated by venous sinusoids, which anastomose with each other via gaps in the plates (**Fig. 67.20**; see Fig. 67.22). Until about 7 years of age, plates are normally two cells thick. This pattern recurs in liver regeneration, when a mixture of hyperplasia and hypertrophy results in cell plates that are several cells thick and composed of larger multinucleated hepatocytes, reverting over the course of about a year to single cell thickness.

Bile is secreted by hepatocytes and collected in a network of minute tubes (canaliculi). The hepatocytes can therefore be regarded as exocrine cells, secreting bile into the alimentary tract via the extrahepatic bile ducts. Their other metabolic functions involve complex biochemical exchanges with the blood.

The fetal liver is a major haemopoietic organ; erythrocytes, leukocytes and platelets develop from mesenchyme covering the sinusoidal endothelium.

## Lobulation of the liver

The structural unit of the liver is the lobule: a roughly hexagonal arrangement of plates of hepatocytes, separated by intervening sinusoids that radiate outwards from a central vein, with portal triads at the vertices of each hexagon (**Fig. 67.21**; see Fig. 67.19). The central veins drain into the hepatic veins. In some species, the classic lobular units are delimited microscopically by distinct connective tissue septa. However, the lobular organization of the human liver is not immediately evident in histological sections; the lobules do not have distinct boundaries, and connective tissue is sparse. The plates do not pass straight to the periphery of a lobule like the spokes of a wheel but run irregularly as they anastomose and branch.

Detailed studies of human liver, using three-dimensional reconstruction and morphometric analysis combined with histopathological observations, have revealed a highly ordered arrangement of functional units: the hepatic (portal) acini. Each acinus is an approximately ovoid mass of tissue, orientated around a terminal branch of a hepatic arteriole and portal venule, and with its long axis defined by two adjacent central veins (see Fig. 67.19). It includes the hepatic tissue served by these afferent vessels and is bounded by adjacent acini.

The acinar definition of hepatic microarchitecture has clarified the interpretation of liver histopathology, particularly in relation to zones of hypoxic damage, glycogen deposition and removal, and toxic injury, all of which are related to the direction of blood flow. There are also real metabolic differences between hepatocytes within the acini, which can be divided into three zones: zone 1 (periportal) is nearest to the terminal branches of afferent vessels; zone 2 is the intermediate zone; and zone 3 is the area closest to the central vein.

## Blood supply

Preterminal hepatic arterioles within the acini convey arterial blood to the sinusoids, mostly via a fine capillary plexus that drains to branches of the portal veins; a small proportion of arterial blood passes directly to the hepatic sinusoids, bypassing these capillary plexuses. Sinusoids thus contain mixed venous and arterial blood. Central veins from adjacent lobules form interlobular veins that unite as hepatic veins and drain to the inferior vena cava.

## Hepatic plates (cords)

The endothelium of the sinusoids is fenestrated and lacks a basal lamina, which enables it to act as a dynamic blood filter (Fraser et al 1995). The sinusoids are separated from the plates of hepatocytes by a narrow gap, the perisinusoidal space of Disse, which is normally 0.2–0.5 µm wide but which distends in hypoxic states. The space contains interstitial fluid, the microvilli of adjacent hepatocytes, hepatic stellate cells, fine collagen fibres (mostly type III, with some types I and IV), and occasional non-myelinated nerve terminals.

A network of minute, interconnecting biliary canaliculi (approximately 1 µm in diameter) runs between the hepatic plates. The canaliculi are formed by the apposed plasma membranes of adjacent hepatocytes sealed by tight junctions. They conduct bile to the canals of Hering, trough-like structures lined by cholangiocytes and hepatocytes within the liver lobules. Each canal collects bile from multiple canaliculi and empties into a bile ductule (lined by cholangiocytes), which in turn drains into an interlobular bile duct within the portal tracts (Roskams et al 2004). The flow of bile is thus towards the periphery of the lobule, in the opposite direction to the blood flow, which is centripetal.

## Cells of the liver

Resident cells of the liver include hepatocytes, hepatic stellate cells (also known as fat-storing Ito cells), sinusoidal endothelial cells, macrophages (Kupffer cells), the epithelial cells of the biliary tree (cholangiocytes), hepatic stem cells, natural killer lymphocytes (pit cells), and connective tissue cells of the capsule and portal tracts (see Fig. 67.19B).

**Hepatocytes** About 80% of the liver volume and 60% of its cellular population is made up of hepatocytes (parenchymal cells) (see Fig. 67.21; **Fig. 67.22**). They are polyhedral, with 5–12 sides, and measure 20–30 µm across. Their nuclei are round, euchromatic and often tetraploid, polyploid or multiple, with two or more in each cell. Their cytoplasm typically contains a considerable amount of rough and

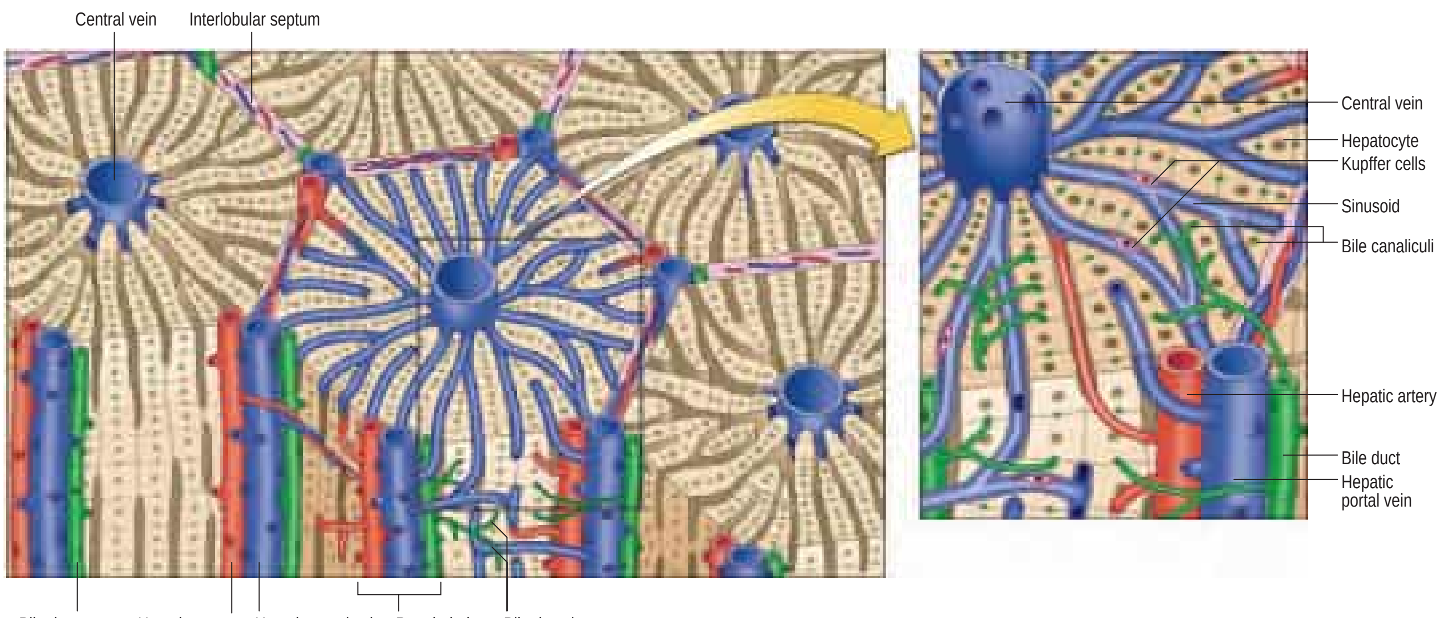



**Fig. 67.21** The hepatic microstructure. Sinusoidal endothelial cells are not shown.

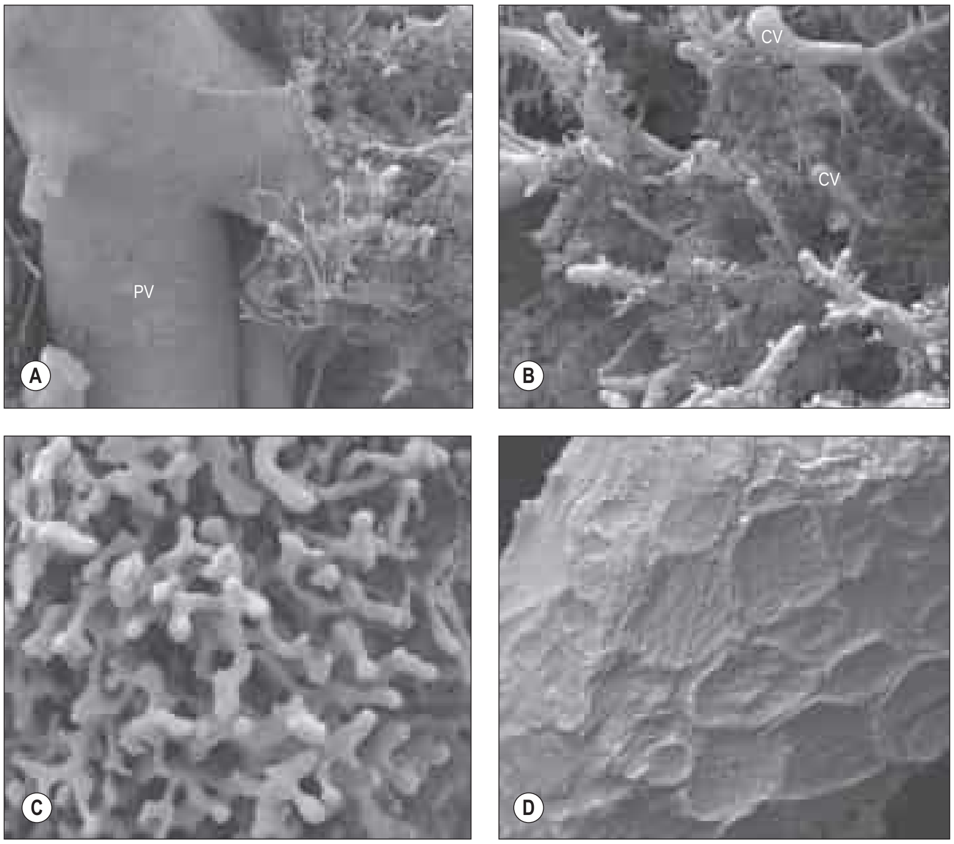



**Fig. 67.20** Scanning electron micrographs of a resin corrosion cast of the human liver demonstrating the three-dimensional arrangement of the hepatic sinusoids. **A**, A portal vein (PV) branch within a portal triad ×33 100 µm. **B**, Sinusoids and central veins (CV) ×37 100 µm. **C**, A sinusoidal network ×330 10 µm. **D**, Sinusoidal endothelial cells × 7000 1 µm. (Courtesy of Professor Greg Jones, University of Otago, New Zealand).

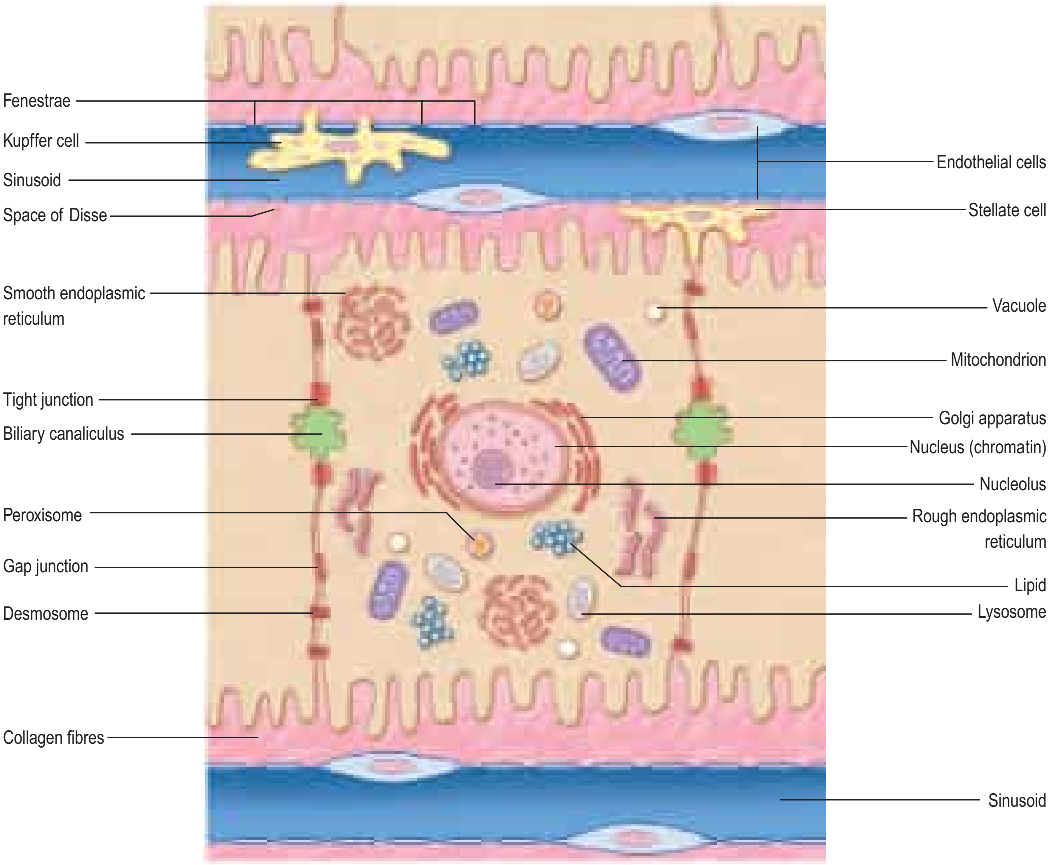



**Fig. 67.22** A schematic illustration of a hepatocyte and adjacent sinusoids. (Redrawn from O'Grady J, Lake JR, Howdle PD. Comprehensive Clinical Hepatology, Mosby 2000.)

smooth endoplasmic reticulum, many mitochondria, lysosomes and well-developed Golgi apparatus, all features that indicate a high metabolic activity. Glycogen granules and lipid vacuoles are usually prominent. Numerous large peroxisomes and vacuoles containing enzymes, e.g. urease in distinctive crystalline forms, indicate the complex metabolism of these cells. Their role in iron metabolism is reflected by the presence of storage vacuoles containing crystals of ferritin and haemosiderin.

The surfaces of hepatocytes facing the sinusoids exhibit numerous microvilli, approximately 0.5 µm long, creating a large surface area (approximately 70% of the hepatocyte surface) exposed to blood plasma. Elsewhere, hepatocytes are linked by numerous gap junctions and desmosomes. Lateral plasma membranes of adjacent hepatocytes form microscopic channels, the bile canaliculi, which are specialized regions of intercellular space formed by apposing grooves in hepatocyte plasma membranes, sealed from extraneous interstitial space by tight junctions. Four types of transmembrane molecules (occludins, claudins, junctional adhesion molecules and coxsackie virus and adenovirus receptors) constitute the core units of the functional protein complexes of the tight junctions (Lee 2012) (see Fig. 1.19). Individual tight junction complexes, either alone or in combination, are involved in signalling pathways in health and disease. Numerous membrane-bound exocytotic vesicles cluster near the lumen of the canaliculi because the secretion of bile is targeted at the canalicular plasma membrane. The tight junctions surrounding the biliary canaliculi prevent bile from entering the interstitial fluid or blood plasma, thereby creating a blood–bile barrier.

**Hepatic stellate cells** Hepatic stellate cells are much less numerous than hepatocytes. They are irregular in outline and lie within the perisinusoidal space of Disse between the sinusoids and the hepatocyte plates. They are thought to be mesenchymal in origin and are characterized by numerous cytoplasmic lipid droplets. These cells secrete most of the intralobular matrix components, including collagen type III (reticular) fibres. They store the fat-soluble vitamin A in their lipid droplets and are a significant source of growth factors involved in liver homeostasis and regeneration. Hepatic stellate cells also play a major role in pathological processes (Yin et al 2013). In response to liver damage, they become activated and predominantly myofibroblast-like. They are responsible for the replacement of damaged hepatocytes with collagenous scar tissue, a process called hepatic fibrosis, that is seen initially in zone 3, around central veins. Fibrosis can progress to cirrhosis, where the parenchymal architecture and pattern of blood flow within the liver are destroyed, with major systemic consequences.

**Sinusoidal endothelial cells** Hepatic venous sinusoids are generally wider than blood capillaries and are lined by a thin but highly fenestrated endothelium that lacks a basal lamina. The endothelial cells are typically flattened, each with a central nucleus, and are joined to each other by junctional complexes. The fenestrae are grouped in clusters with a mean diameter of 100 nm, allowing plasma direct access to the basal plasma membranes of hepatocytes. Their cytoplasm contains numerous transcytotic vesicles.

**Kupffer cells** Kupffer cells are hepatic macrophages derived from circulating blood monocytes and originate in the bone marrow. They are long-term hepatic residents and lie within the sinusoidal lumen attached to the endothelial surface. Kupffer cells are irregular in shape and have long processes that extend into the sinusoidal lumen. They form a major part of the mononuclear phagocyte system, which is responsible for removing cellular and microbial debris from the circulation, and for secreting cytokines involved in defence. They remove aged and damaged red cells from the hepatic circulation, a function normally shared with the spleen but fulfilled entirely by the liver after splenectomy.

**Hepatic stem cells** Hepatic stem cells are undifferentiated, pluripotential cells and are understood to reside around the canals of Hering; they are derived from the ductal plate in fetal livers. Their progeny are small epithelial cells (8–18 µm) with a large oval nucleus and scanty cytoplasm, expressing both biliary epithelial and hepatocyte markers (Roskams et al 2010).

## Bonus e-book images

**Fig. 67.10** Common hepatic artery variants.

**Fig. 67.11** Hepatic arteriograms.

**Fig. 67.16 A**, The umbilical fissure that divides the anterior sector of the left lobe into two segments is, in fact, an artificial scissure. **B**, When it is suppressed, the anterior sector appears as a single segment.

**Fig. 67.17** The liver is divided into three segments and a caudate area, according to the ramification of the Glissonian pedicles.

**Fig. 67.18** Resectional terminology.

**Fig. 67.20** Scanning electron micrographs of a resin corrosion cast of the human liver demonstrating the three-dimensional arrangement of the hepatic sinusoids.

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